



Review

Digital image steganography survey and investigation (goal, assessment, method, development, and dataset)



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ABSTRACT

Digital steganography has a long history, starting to be developed in the 90s until now. The main aspects of early steganography are security, imperceptibility, and payload. Security is an essential aspect of steganography, where its development competes with the steganalysis method. The development of steganographic methods is not only based on statistics but also involves adaptive methods, machine learning, artificial intelligence, generative adversarial networks (GAN), convolution neural networks (CNN), coverless techniques, etc. Several new methods add robustness as an important aspect. This makes the development of steganographic methods more varied and generally focuses the goal of the method on one or two aspects. Nowadays, steganography articles, especially surveys, rarely explicitly classify steganography based on its goals. So this survey contributes to classifying steganographic research based on goals and its assessments. This paper also reviews the use of assessment tools in depth because it is closely related to the goal of steganography. Challenges or issues, methods, developments, and popular datasets are also reviewed. Steganalysis literature was also added to improve the quality of the study and analysis. Finally, this survey provides discussion, investigation, critical analysis, and explicit summary so novice researchers can more easily understand image steganography development.

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1. Introduction

1.1. Information Security and Data Hiding

The landscape of human life is increasingly shifting to the digital world. Today's digitalization does not only involve private networks but involves more public networks. The need for digital data and information transactions makes cybercrime also growing. So, security methods also need to be developed to protect transactions and information storage from attacks by hackers. Cryptography and data hiding are methods commonly used to protect confidential data and information. But the two are different disciplines. Data hiding has two main scientific branches, namely, steganography and watermarking. Cryptography, steganography, and watermarking have their own goals and characteristics. Steganography is one method of securing data by hiding secret data into a media cover to trick enemies or hackers. The emphasis of steganography is securing hidden secret data. Watermarking is a method of em-

bedding a watermark into the media cover to protect the ownership of the media cover. However, in the early digital data hiding development era, the difference between steganography and watermarking was insignificant. Because at the beginning of the development of the watermarking method, the level of robustness was not significant, as was the level of security of steganography. While cryptography is a security method carried out by transforming the form of secret data so that it cannot be read and even looks damaged [1–3]. To know more clearly the general differences between steganography, watermarking, and cryptography, see Table 1.

Steganography is a technology that has been utilized for thousands of years and has been used for thousands of years. Steganography comes from the Greek words “stegos” and “grafia” which means “covered writing” [2,4]. Contemporarily one of the reasons why it is necessary to hide messages has been explained in Simmons [5]. One of the revolutions of the modern steganographic method at the beginning of the digitalization era can be attributed to the method proposed by Kurak and McHugh [6]. It can be concluded that the beginning of the idea of steganography is based on the least significant bit (LSB) by using the image as a cover. But various steganographic reports can also use various covers such as audio, hyper text markup language (HTML), executable files (.EXE), and extensible markup language (XML) [7,8]. In the digital

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Table 1
Characteristics of Information Security Method [2,19].

Properties/Methods	Information Security		
	Data Hiding		Cryptography
	Steganography	Watermarking	
Aim	Protect the secret data	Protect the cover media	Protect the secret data
Cover Types	Commonly multimedia data, but sometimes on text	Commonly multimedia data	Any media
Cover Selection	Any	Restricted to media to be protected	N/A
Recovery Method	Blind	Blind / Non-blind (using original watermark/cover media)	Blind
Recovery level	Must be perfect	Doesn't have to be perfect but identifiable	Must be perfect
Key	Optional	Optional	Always
Visibility	Invisible	Invisible / Visible	Visible
Important aspects	Security/Undetectability, payload and Imperceptibility	Robustness and Imperceptibility (for invisible watermarking)	Security
Failure if	Secret data detected	Watermark removed or replaced	Secret data deciphered

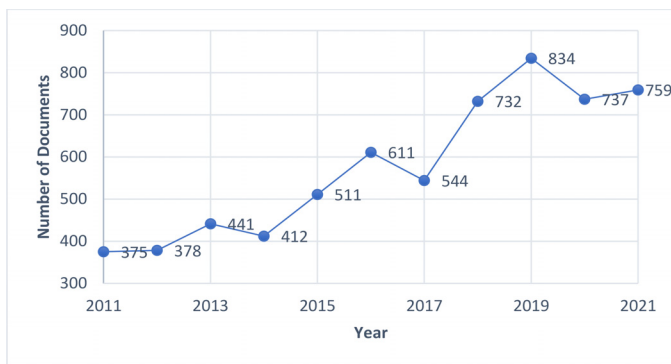


Fig. 1. Document search results with the keyword “image steganography” OR “data hiding” on scopus.com.

era, steganography has been implemented in various ways, such as military, medical, and biometric data, telecommunication protocols, captcha security, QR code for voting, website applications, digital forensics, antivirus, social media, etc [9–18]. In real terms, steganography can be done with good intentions or vice versa for abuse by parties who are not responsible for cyber-attacks. So, the science of steganography needs to be studied to protect data and prevent harmful cyber-attacks.

Based on the data presented in Table 1, it is clear that, in general, these three methods have differences, but in their development, they can intersect, hybrid, or be combined. Additionally, research on image steganography is increasing every year, and this is evidenced by the search results on the scopus.com page for the last ten years with the keywords “image steganography” OR “data hiding” in the title-abstract-keyword fields, see Fig. 1. Several survey papers on popular steganography that have been published previously have summarized, discussed and analyzed it. Survey [2] has discussed the history of steganography from the ancient to the digital era, its difference with watermarking and cryptography, applications of steganography, methods of steganography, analysis, and future recommendations. The survey is very comprehensive, but the steganographic method continues to grow very significantly after a decade of the survey being made. Another survey was conducted [4] focusing on general steganography, fundamental concepts, measuring tools, development of methods used in the spatial domain, and transformation. The last publication discussed was [4] in 2013, so it needs to be updated.

Along with the increasing number of steganographic methods, papers [20] conducted surveys that were more focused on spatial domain steganography. The survey paper [19] is one of the most

recent surveys comprehensively discussing general steganography. The discussion on the steganographic method is very detailed, with an explanation of the algorithm step by step. However, the discussion of the survey [19] is still general. Based on Table 1 and Fig. 1 shows the closeness of the steganography and watermarking methods as derivatives of data hiding. In addition, there are also many steganographic methods combined with cryptography and its derivatives. The development of steganography methods has been very frontal in recent years, but there are some things that were missed in previous survey papers. This is evidenced by the discontinuity between the purpose of steganography and the assessment tools used in several studies discussed in this survey. So a critical analysis is needed and can classify steganographic research more clearly so that the direction of developing steganographic methods can be more focused. Starting from this problem, this survey was conducted to provide a deeper and more explicit perspective. Specifically to investigate and analyze various steganographic methods and their development. Each steganographic method is developed with different approaches, methods, datasets, etc. This difference will undoubtedly determine the assessment instrument used, and if the assessment made are not appropriate, the contribution of steganographic research will be less appropriate. In detail, this survey aims to:

1. Identify the challenges in image steganography and the ideal assessment tool.
2. Identify and analyze state-of-the-art image steganography, then classify types of image steganography based on goals, domains, and approaches.
3. Identify and analyze the development of derivative and sophisticated steganography.
4. Identify datasets that are widely used in image steganography research.

This survey is limited to steganography which uses digital images as cover media. In addition, it does not discuss computational time and complexity because not many steganographic methods include this. Besides that, it is difficult to make a fair comparison, especially for the computational time, due to differences in hardware and software. Several papers on steganalysis are also used for this survey to obtain more valid and data-rich analysis results. Steganalysis is a method to detect whether an image is clean or embedded with a secret message. This paper is organized into several sections, including the introduction. Section 2 discusses various goals and their measurement tools in image steganography. Section 3 discusses steganographic methods and their classification. Some of the latest data hiding methods and the development of image steganography are described and briefly analyzed

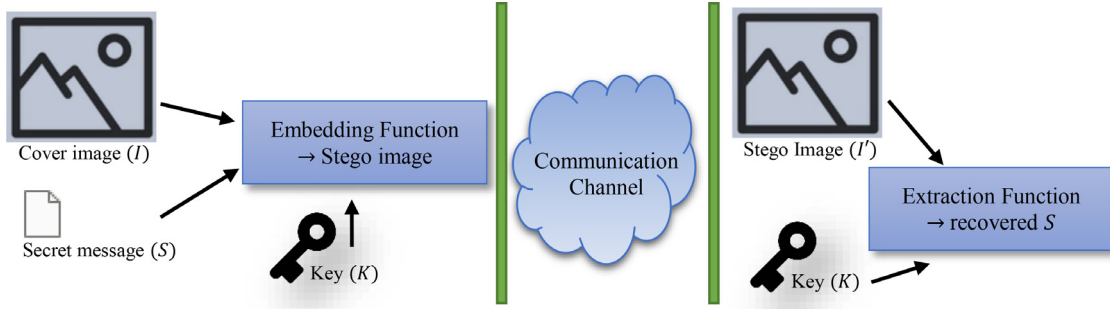


Fig. 2. Generic procedure in Image Steganography.

in section 4. The steganographic image dataset and summary are discussed in sections 5 and 6, respectively.

1.2. Generic Concept and Procedure of Steganography

In general, the concept of steganography is to hide secret messages by using cover carrier media, which aims to trick hackers, so that secret messages are not detected. Steganography is done through two main procedures, namely embedding or, in other terms, sometimes called the coding phase, and extraction or sometimes called the decoding phase. Embedding (Em) and extraction (Ex) processes can be defined in Eq. (1) and Eq. (2), respectively.

$$Em = I \oplus K \oplus S \rightarrow I' \quad (1)$$

$$\therefore Ex(Em(i, k, s)) \approx m, \forall i \in I, k \in K, s \in S \quad (2)$$

Where I is the cover carrier image, S is the secret message, K is the optional key. After the encoding phase, the message is sent through the communication channel to the recipient. On the receiving side, the message is read with a decoding phase. More clearly, the general concepts and procedures of steganography are presented in Fig. 2.

2. Goal in Image Steganography

In data hiding, four main aspects are always measured: imperceptibility, payload capacity, security, and robustness. These four things are simultaneously the goals and challenges of data hiding research [1,2]. But in image steganography, the robustness aspect is considered unimportant because the main purpose of steganography is to avoid detecting the presence of messages [1,2]. However, several recent studies' steganography method requires robustness when passing through a lossy or noisy channel. Robustness here means the resilience of the message to remain in the cover image when an attack or manipulation occurs. The robustness aspect is actually more important in the watermarking method [1,2] because watermarking aims to provide copyright protection except for fragile watermarking, which does not pay attention to the robustness aspect because fragile watermarking works to protect the cover image differently, even self-recovery [21–23]. Image steganography is generally fragile and is mostly done in the spatial domain. In addition, it can be seen again in Table 1, that in generic steganography, the message must be done by blind extraction, and the message should be extracted perfectly so that the message can be conveyed to the receiver.

Imperceptibility is an aspect used to determine the effect of distortion in the cover image after the message is embedded. A too-large distortion can reduce image quality and may even be detectable by human vision [11]. Payload capacity means the size of the secret message that can be embedded in the cover image. The

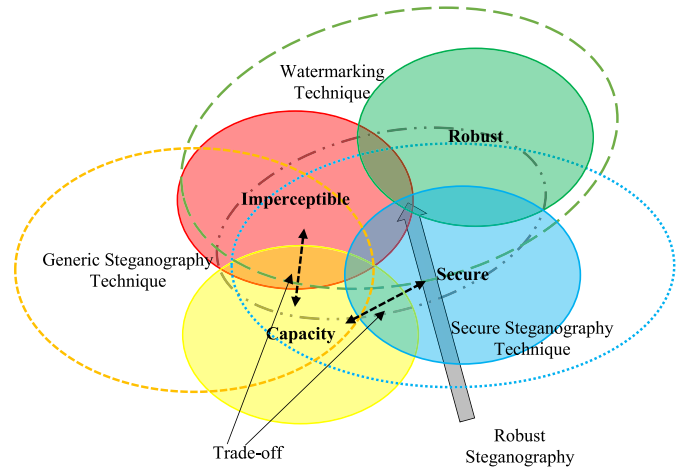


Fig. 3. Generic trade-off and relationship of four aspects in Data Hiding.

greater the payload capacity is more efficient the steganography method because it does not require many covers to send messages. But generally, the bigger the payload, the bigger the impact on the imperceptibility quality [24]. In image watermarking, the payload capacity aspect is generally unimportant [2]. The existence of trade-off aspects of imperceptibility and payload capacity makes it one of the challenges or issues in image steganography research.

Security is an important aspect of image steganography. Secure means that the message is not only imperceptible but must be secure/undetectable. In image steganography research such as [25–27] which aims to reduce the trade-off between imperceptibility and payload (generic steganography technique), sometimes the security aspect is not discussed much. In fact, if a new steganography method is proposed, then at least it needs to be tested for security with standard statistical analysis. The security aspect is the main aspect of steganography because the main purpose of steganography is to secure the delivery of messages on public networks. Generally, this aspect also has a trade-off in capacity. With the larger capacity, the message will be easier to detect. This is also one of the exciting challenges in the development of secure steganography research [28]. However, there has not been much steganographic research that explicitly discusses if a secret message has been detected and whether the detected message can be extracted perfectly by the attacker or just thwarted its delivery. Furthermore, each aspect is discussed in more detail in sub-chapters 2.1 to 2.4. While Fig. 3 describes the relationship between these four aspects, which this relation we conclude the survey of all references, especially in Table 6.

2.1. Imperceptibility and its assessment

Imperceptibility generally can be calculated by comparing the pixel values between the original cover and the stego image (reference measurement). Still, some measuring instruments can perform calculations only with the stego image (no reference measurement). In more detail, the following are tools often used to measure imperceptibility.

2.1.1. Mean Square Error (MSE) and Root Mean Square Error (RMSE)

MSE is a tool to measure the level of squared error that occurs in stego image pixels. MSE is a measuring tool that requires a reference. The MSE value can be calculated by Eq. (3).

$$MSE = \frac{1}{WH} \sum_{x=1}^W \sum_{y=1}^H (I_{xy} - I'_{xy})^2 \quad (3)$$

Where W is width, H is height, x, y is coordinates of the image cover. Please note that the MSE value is only calculated for one image layer (grayscale image). To calculate the color image, it is necessary to add one more summation to the image layer (L), see Eq. (4).

$$MSE = \frac{1}{WH} \sum_{x=1}^W \sum_{y=1}^H \sum_{z=1}^L (I_{xyz} - I'_{xyz})^2 \quad (4)$$

Another form of MSE is RMSE. Because in MSE there is a quadratic process, then in RMSE a square root is carried out. Thus, a smaller value will be obtained. However, this value refers to the same value as the MSE value. RMSE can be calculated by Eq. (5).

$$RMSE = \sqrt{MSE} \quad (5)$$

2.1.2. Peak Signal-to-Noise Ratio (PSNR)

PSNR is the most widely used, popular, and long-standing imperceptibility measurement tool. PSNR can be known as the noise level in stego image pixels. PSNR has units of decibels (dB), and when perfect results are obtained, $PSNR = \infty$. In generic steganography, this is not possible, and this is due to the distortion that occurs in the image due to embedding. The greater the payload capacity, the PSNR value will decrease. Stego images with excellent quality have at least a PSNR value of 40 dB or better, while when the value is less than 30 dB, the quality tends to be less good [2]. PSNR is calculated based on the logarithmic MSE value, see Eq. (6), or with RMSE with Eq. (7).

$$PSNR = 10 \log 10 \left(\frac{I_{max}^2}{MSE} \right) \quad (6)$$

$$PSNR = 20 \log 10 \left(\frac{I_{max}}{RMSE} \right) \quad (7)$$

Where I_{max}^2 is the maximum pixel value of the image. I_{max}^2 the largest is 255 for grayscale images (2^8), for binary image I_{max}^2 he largest is 1. In some research [25,29–68], $C_{max}^2 = 255$ is used as the default value. Whereas in certain cases it is possible that the value of C_{max}^2 does not reach 255, when this occurs and the value remains $I_{max}^2 = 255$ then the accuracy of the PSNR measurement is less valid.

2.1.3. Structural Similarity Index (SSIM)

MSE and PSNR have already recognized measuring tools in various digital image processing. But it is not enough because it cannot measure the fidelity of an image. SSIM is a newer measurement tool than PSNR, first published in Wang et al. [69]. SSIM is designed based on three primary parameters: local luminance and contrast and image structure independent of luminance and contrast [69]. SSIM is an image quality measurement tool with a

paradigm based on the human visual system (HVS). SSIM is also considered more sensitive to histogram analysis in testing several basic steganographic methods. In research [24], it has been proven that the measurements of SSIM and PSNR are not always appropriate. When the PSNR value is better, it is not necessarily the SSIM value that is also better. Although the PSNR has a very good value, when the SSIM value is not good enough, there can be a change in the shape of the histogram. But when the SSIM value is excellent, the image structure based on histogram analysis is more similar to the cover image. The SSIM value is generally between 0 and 1, where a value close to 1 indicates better SSIM quality. But in rare cases, the SSIM value can also be close to or equal to -1. As a note, although the difference in SSIM values is very small (four digits behind the comma), when measured by PSNR, the difference can be quite large (>1 dB). Specifically, the SSIM between the cover image (I) and the stego image (I') can be calculated by Eq. (8).

$$SSIM(I, I') = \frac{(2\mu_I \mu_{I'} + C_1)(2\sigma_{II'} + C_2)}{(\mu_I^2 + \mu_{I'}^2 + C_1)(\sigma_I^2 + \sigma_{I'}^2 + C_2)} \quad (8)$$

$$C_1 = (sc_1 D)^2 \quad (9)$$

$$C_2 = (sc_2 D)^2 \quad (10)$$

Where μ is the mean intensity of luminance, σ standard deviation of signal contrast, sc_1 and sc_2 are stabilizing parameter which contains small constant value ($\ll 1$), by default sc_1 and sc_2 are 0.01 and 0.03, and D is the dynamic range of pixel value (255 for 2^8 grayscale image).

2.1.4. Mean SSIM (MSSIM) and Multi-scale SSIM (MS-SSIM)

MSSIM is a form of measuring SSIM on the whole image. MSSIM is calculated from the average value of the local image window [69]. In image steganography, MSSIM is not more popular than SSIM. However, MSSIM is needed to measure image quality as a whole. Based on tests in Mittal et al. [70], it is proven that MSSIM has better performance than SSIM, but the measurements are only limited to natural images and not specific for steganographic images but universally for various digital image processing. MSSIM can be calculated by Eq. (11).

$$MSSIM(I, I') = \frac{1}{M} \sum_{x=1}^M SSIM(i_x, i'_x) \quad (11)$$

Where M is the number of local windows of image, i and i' are the local window of images I and I' , and x is the local index window.

MS-SSIM is a multi-scale approach to SSIM. In various research, this measuring instrument steganography image is the least common compared to SSIM and MSSIM. This is also because this measuring instrument needs to be developed again to make it more stable. Some research that uses MS-SSIM is research [71]. In MS-SSIM, a low pass filter is applied, reducing the sample input signal based on a factor of two iteratively. MS-SSIM provides more flexibility than a single scale in combining viewing conditions and variations in image resolution. Unfortunately, this method is still a bit rough. The original image signal is indexed on a scale of 1, and the largest scale is the M scale, with a number of iterations of $M-1$. MS-SSIM can be calculated by Eq. (12) [72].

$$MS-SSIM(I, I') = [l_M(I, I')]^{\alpha_M} \cdot \prod_{x=1}^M [c_x(I, I')]^{\beta_x} [s_x(I, I')]^{\gamma_x} \quad (12)$$

Where l is luminance, c is contrast, s is structure, while α_M , β_x , and γ_x are the parameters used to adjust the relative importance of the different components.

2.1.5. Universal Image Quality Index (UIQI)

UIQI is an image quality measurement method used in image steganography. Like MSE, PSNR, SSIM, MSSIM, and MS-SSIM, UIQI is a measuring tool that requires a reference. UIQI was first proposed by [73], but it is also often abbreviated as Quality Index (QI) in various studies [43,74]. The UIQI method is not better than SSIM because SSIM is a refinement of UIQI. UIQI is the first measuring instrument designed based on the characteristics of the human visual system by combining three main factors: contrast distortion, loss of correlation, and luminance. The UIQI value ranges from -1 to 1, where the best value is 1. The UIQI can be calculated by Eq. (13).

$$Q = \frac{4\sigma_{II'}}{(\sigma_I^2 + \sigma_{I'}^2) \left[(\bar{I}^2) + (\bar{I'}^2) \right]} \quad (13)$$

Where $\bar{I}$ and $\bar{I'}$ is mean intensity, $\sigma_{II'}$ is the covariance of both image, $\sigma_{I'}$ and σ_I are standard deviation of stego and cover image, respectively [75].

2.1.6. Feature Similarity Index (FSIM)

FSIM is one of the full reference measuring tools published by [76]. In image steganography research, FSIM is used in [32,77]. Just like SSIM and UIQI, FSIM is built on the characteristics of HVS. The fact that HVS understands its primary image from low-level features. The main feature used in FSIM is phase congruency (PC). In addition, there is also a secondary feature, namely gradient magnitude. FSIM works with two levels of computational processes, at the first level, a local similarity map is calculated. A pool similarity map is performed on a single similarity score at the second level. FSIM is used to measure grayscale image quality and can be calculated by Eq. (14).

$$FSIM = \frac{\sum_{x \in \Omega} S_L(x) \cdot PC_m(x)}{\sum_{x \in \Omega} PC_m(x)} \quad (14)$$

Where Ω is the whole image, S_L is the overall luminance similarity between the cover and stego image, x is the index.

FSIM cannot be used to measure color image quality because the calculation is intended for grayscale or color image luminance. To measure the quality of color images, RGB images need to be converted into YIQ images. The goal is to calculate the quality on the chrominance channel. The FSIM for measuring color image quality is called $FSIM_C$, which can be calculated by Eq. (15).

$$FSIM_C = \frac{\sum_{x \in \Omega} S_c(x) \cdot [S_c(x)]^\lambda \cdot PC_m(x)}{\sum_{x \in \Omega} PC_m(x)} \quad (15)$$

Where S_c is the overall chrominance similarity between the cover and stego image, λ is the parameter that controls the prominence of the chromatic components.

2.1.7. Image Fidelity (IF)

Another measuring tool used in image steganography is IF. IF is used in research [32,44] and has been surveyed in the study [4,20]. But not much information about this method. But if you look at Eq. (16), the IF measurement is only a measurement based on the number of errors in the pixel value, not much different from MSE and has a value range of 0 to 1.

$$IF = 1 - \frac{\sum_{x=1}^W \sum_{y=1}^H (I_{xy} - I'_{xy})^2}{\sum_{x=1}^W \sum_{y=1}^H (I_{xy})^2} \quad (16)$$

2.1.8. Weighted Peak Signal-To-Noise Ratio (WPSNR)

WPSNR is a measuring tool developed from PSNR with weighting [78]. The goal is to provide measurements that are not only

based on the error but also reflect human perception. WPSNR calculation can be calculated using Eq. (17). In the WPSNR steganography image used in research [79], but not widely used and not as popular as PSNR and SSIM. Although WPSNR is claimed to have good performance for measuring image quality, it is accompanied by inexpensive computations [78].

$$WPSNR = 20 \log_{10} \left(\frac{I_{max}}{\sqrt{MSE} \times NVF} \right) \quad (17)$$

$$NVF = \frac{1}{1 + \theta \sigma_I^2} \quad (18)$$

Where NVF stands for noise visibility function which is a texture masking function that acts as a penalty factor, θ is adjustable parameter, σ_I^2 is local variance image with 3×3 window.

2.1.9. Kullback-Leibler (KL) Divergence

KL Divergence, often referred to as relative entropy, is a measuring tool that was first published in [80]. KL divergence is a calculation tool to measure the statistical distance between the probability distribution of two images. The embedding process in steganography will cause distortion, resulting in variations in the histogram distribution. Thus, KL divergence can measure the divergence between the cover and stego images [81,82]. Actually, KL divergence can be said to be a measure of imperceptibility and security. Because when the value is getting smaller or closer to 0, it indicates that the distortion is getting smaller, and the histogram distribution will approach the cover image, so the possibility of the message being detected is slighter [83]. KL divergence can be calculated by Eq. (19).

$$D_{KL}(I||I') = \sum_{x \in X} I_x \log \frac{I_x}{I'_x} \quad (19)$$

2.1.10. Visual Information Fidelity (VIF)

VIF is a measuring tool that was first published in [84]. VIF is designed by modeling the physiological and psycho-visual features that stand out from the characteristics of the human visual system. VIF is designed based on natural image more 'consumed' by humans, so it has not been widely proven in other images such as medical or graphics. In image steganography, VIF generally has a range of 0 to 1, but the VIF value can be more than 1 when the stego image is considered to have better visualization [24]. The use of VIF in image steganography is not as popular as SSIM or UIQI. Some image steganography research using the VIF are [55,83]. Still, this measuring tool is certainly more recommended when used in image steganography research combined with image enhancement, for example, in research [85]. This is used to determine whether the image quality after embedding is better. VIF can be calculated by Eq. (20).

$$VIF = \frac{\sum_{j \in \text{subbands}} I(\bar{C}^N, j; \bar{F}^N, j | S^N, j)}{\sum_{j \in \text{subbands}} I(\bar{C}^N, j; \bar{E}^N, j | S^N, j)} \quad (20)$$

$$I(\bar{C}^N; \bar{F}^N | S^N) = \frac{1}{2} \sum_{x=1}^W \sum_{y=1}^H \log_2 \left(1 + \frac{g_x^2 s_x^2 \lambda_y}{\sigma_v^2 + \sigma_n^2} \right) \quad (21)$$

$$I(\bar{C}^N; \bar{E}^N | S^N) = \frac{1}{2} \sum_{x=1}^W \sum_{y=1}^H \log_2 \left(1 + \frac{s_x^2 \lambda_y}{\sigma_n^2} \right) \quad (22)$$

Where $I(\bar{C}^N; \bar{F}^N | S^N)$ and $I(\bar{C}^N; \bar{E}^N | S^N)$ is reflect the information that the brain might ideally retrieve from a certain subband in the reference and test image, respectively, k is the index of subbands, σ_n^2 is the variance of visual noise, g_x is attenuation factors denote a realization for a particular reference image, λ_y is eigen value, s_x denotes a realization for a particular reference image.

2.1.11. BRISQUE score

BRISQUE score is a natural scene statistical measurement tool without reference [70]. This differs from PSNR and SSIM, which require the original image as a reference. We conclude that the word “natural” in natural scene statistics refers to natural images. BRISQUE scores in the 30–40 range have better perceptual quality [42]. However, theoretically, BRISQUE scores range from 0 to 100, where the best score is the lowest [85]. This value will be obtained in images close to the original or with the least distortion. The BRISQUE score is calculated using the support vector regression (SVR) model trained on the image database, and the accompanying differential mean opinion score (DMOS) [70]. In image steganography research, the BRISQUE score has been used in research [42,85].

2.1.12. Mean Opinion Score (MOS)

MOS is a subjective image quality measuring tool. This is different from all the measuring instruments that have been discussed previously. To determine the MOS, several observers are needed to assess the quality of the image, where each observer has the expertise and, of course, can give different values. Generally, MOS values are between 1 and 5, where five is the best and one is the worst [86]. MOS is rarely used in various image steganography research and image processing research because it requires expert observers. However, MOS is mostly used to test various image quality measurement tools that have been discussed previously. However, in research [87] MOS is used as a measuring tool that involves ten observers, consisting of five men and women, each who are Ph.D students and post-doc fellows.

2.1.13. Summary of Imperceptibility Assessment

Many imperceptibility assessment tools have been used in image steganography research. From the thirteen measuring instruments described, it can be concluded that MSE, RMSE, and PSNR are measuring instruments based on error calculations. Where PSNR and MSE are the most widely used and well-accepted measuring tools. However, assessment tools based on error calculations are still considered lacking. At least one or two other assessment tools designed based on the characteristics of HVS should be added, such as WPSNR, SSIM, MSSIM, MS-SSIM, UIQI, and FSIM. SSIM is a measuring tool that is considered the most stable and is widely used in image steganography research. SSIM is also claimed to be better than UIQI. However, SSIM is calculated based on local features, so MSSIM can also be used as an option to assess the overall image quality. MS-SSIM is still “rough” so it is considered unstable and not widely used. While WPSNR and FSIM are not very popular. WPSNR tends to be the simplest because it is developed from PSNR with added NVF parameters, so it does not require expensive computations. Meanwhile, IF does not have much information, even though the name shows the impression of HVS characteristics. Judging from the formula, it looks like an error calculation, so we do not recommend this measuring tool. KL divergence is a different measuring tool, here, it is not only measures imperceptibility but also can measure the level of security based on variations in the histogram distribution. BRISQUE and VIF are different measuring tools and are designed for natural images. VIF can measure the similarity with the original image and the better quality of the original image. If the other is a full reference measuring tool, BRISQUE is a no-reference measurement tool. BRISQUE and VIF can be used for additional purposes, such as measuring the quality of image steganography combined with image enhancement. Meanwhile, MOS will be quite difficult to use because the assessment is subjective and depends on the observer's expertise. Some other measuring tools that are also used in image steganography are structural content (SC) [66,83], pearson correlation coefficient (PCC) [55,83], normalized coefficient [32],

normalized cross coefficient (NCC) [36,41,56,75,87–91], etc. However, correlation coefficient-based measuring tools such as PCC, NC, and NCC is more suitable for measuring the message extraction process, especially for image-based messages [2]. For the record, some researchers sometimes understand that imperceptibility is the same as security in research [92]. Although it certainly has a direct relationship, what is meant by imperceptibility is a message that cannot be felt by the human senses, while security is an undetectability stego image, which is more clearly described in section 2.3. We summarize the use of imperceptibility measurement tools in Table 2 based on the research described in Table 6.

2.2. Payload Capacity and its assessment

2.2.1. Payload Capacity (PC)

Payload capacity in steganographic images commonly is measured in bits. This is because one image pixel consists of one byte for grayscale images and three or four bytes for color images. One pixel color image consists of several channels, where each channel consists of one byte, while the grayscale image only consists of one channel. The embedding for each pixel in each channel is theoretically always smaller than the pixel size. PC is obtained by counting all the bits embedded in an image, but research also writes with byte sizes such as [34,97]. Some other terms of payload capacity are capacity, embedding capacity, hiding capacity, or bits capacity. The PC will be significantly influenced by the size of the cover image and the method used, but with a small cover size, you must have a small PC too. Therefore, the BPP measure (in section 2.2.2) is also important to include to support the presentation of the results.

2.2.2. Bit per Pixel (BPP)

BPP refers to the average value of embedding per pixel of the whole image. Another term that has the same meaning as BPP is bits per byte (BPB), as used in Swain [100], and bits per channel (BPC), as in research [123]. BPP can be calculated by PC divided by the number of pixels in the image, see Eq. (21). BPP has a more general size and does not depend on image size because BPP is the average embedding size of each pixel. The BPP unit can be used in the spatial and frequency domains.

$$BPP = \frac{PC}{W \times H \times L} \quad (23)$$

2.2.3. Bits Per non-zero DCT Coefficient (bpc) and Bits per Non-zero AC (bpnzAC)

Bits per non-zero AC (bpnzAC) is a capacity measurement tool similar to BPP. Still, bpnzAC is specialized in JPEG steganography, specifically by using a domain transformation, namely the discrete cosine transform (DCT). AC is the coefficient of the DCT transformation. While the bpc measuring tool means bits per non-zero of all DCT coefficients.

2.2.4. Summary of Payload Capacity Assessment

The challenge in generic image steganography research is increasing capacity while maintaining imperceptibility quality. In the spatial domain, generally the payload capacity is more than 1BPP. In standard methods such as the least significant bit (LSB) substitution, the embedded message is at least 1 bit for each pixel by default. If all pixels in an image are embedded with one message bit, then the capacity is 1BPP [24]. Another standard method, pixel value differencing (PVD) can accommodate an average capacity of around 1.5BPP. The current development is very advanced. It can accommodate more than 3 BPP with PSNR quality >30dB and SSIM 0.9. However, in the secure and robust steganography method, the payload size is relatively small, only around 0.05 to

Table 2
Imperceptibility Assessment Tools in Image Steganography Research in Recent Years.

Measurement Tools	Researches
MSE	[25,26,36,37,41,42,44–48,55,56,61,63,65,75,77,79,81,83,87,88,90,92–104]
RMSE	[32,99,105]
PSNR	[24–27,32,34–42,44–61,63,65,66,68,71,74,75,77,79,81,83,85,87–121]
WPSNR	[79]
SSIM	[24,32,35–39,41,42,46–48,53,54,57,58,60,61,63,71,75,77,79,83,85,87,90–95,97,99,102,104,105,108,109,111,114,116,117,122]
MSSIM	[32,89,103]
MS-SSIM	[71]
UIQI	[39,44,47,49,63,68,74,75,91,100,109,114,117]
FSIM	[32,77]
VIF	[55,83]
KL Divergence	[41,56,66,81–83]
BRISQUE score	[42,85]
IF	[32,44,66,89]
MOS	[87]

Table 3
Map of the use of Payload Capacity Assessment Tools in Image Steganography research in Recent Years.

Assessment Tools	Researches
Capacity (bits)	[26,31,33–35,38–40,44,47–49,51,52,54,57,58,60,61,63,65,68,74,82,90,92,97, 98,100,102,103,107,109,114,118,122]
BPP	[25–27,29–31,34,35,37–42,45–47,49–51,53,54,58–60,74,75,77,81,83,85,88,91,93,94,97,100,106,109, 113,116,124–126]
Payload / Capacity in percentage	[33,48,56,108]
bpnzAC	[110,111,115,119,120,127–137]

0.5 BPP/bpnzAC. Table 3 summarises the use of payload measurement tools in the research described in Table 6.

2.3. Security and its assessment

Steganography security can be measured with steganalysis tools. Departing from visual analysis by human eyes is the simplest steganalysis [11]. However, this will certainly be very difficult to do considering the development of increasingly sophisticated steganographic methods. The stego images that are difficult to distinguish without statistical tools. Visual analysis precisely leads to the imperceptibility measurement discussed in section 2.1. This is the initial requirement that determines the quality of the steganographic method. More specifically, for security measurements, a specific steganalysis method is used to detect the presence of messages. Initially, the steganalysis method was originally designed specifically to detect certain steganographic methods, for example, specifically for LSB, PVD or other methods. Along with the development of steganalysis science, it began to be designed universally. Here are some steganalysis measuring tools widely used in various steganographic methods, which are divided into two categories: specific steganalysis and universal steganalysis.

2.3.1. Specific Steganalysis

2.3.1.1. Histogram Analysis. Histogram analysis is basically the simplest steganographic measuring tool and can be observed visually with the human eye. In traditional methods such as LSB (see Fig. 4(c)), a layperson can see significant histogram changes. However, with the development of the steganographic method, histogram analysis is increasingly difficult to observe by manual visual observation. In Fig. 4(a) and Fig. 4(b), the histogram difference is not very significant, but if you look closely, there are still differences. But with only visual observation, it is difficult to determine objectively which method is the best. So, this steganalysis method needs to be combined with other methods. In addition, the histogram method is difficult to analyze for sophisticated steganographic methods.

2.3.1.2. Chi-square X^2 analysis. The chi-square analysis was designed based on the statistical analysis of Pairs of Values (PoVs)

performed while embedding confidential data. The purpose of this approach is to identify LSB-based embedding [139]. Because on LSB, the message will be distributed uniformly with the reversal of the value at embedding. Thus, the appearance of the two values is more the same than the cover image. The more uniform the message bits are hidden in the cover image, the more time the PoVs appear the same [140]. X^2 can be calculated by Eq. (24).

$$X_{k-1}^2 = \sum \frac{(o_{2x} - o_e)^2}{o_e} \quad (24)$$

$$o_e = \frac{o_{2x} + o_{2x+1}}{2} \quad (25)$$

Where o is the occurrence times of POVs ($2x, 2x + 1$), $k - 1$ is the degree of freedom, $x = 1, 2, \dots, 128$ for the grayscale image. Furthermore, the embedding probability can be calculated by Eq. (26).

$$p = 1 - \frac{1}{2^{\frac{k-1}{2}} \Gamma(\frac{k-1}{2})} \int_0^{X_{k-1}^2} e^{-\frac{\rho}{2}} \rho^{\frac{k-1}{2}-1} d\rho \quad (26)$$

Where Γ is the Euler gamma function

2.3.1.3. Regular Singular (RS) analysis. The RS analysis was first designed by Fridrich et al. [141] to detect the presence or absence of a message in an image. In particular, this method is used to detect stego images created based on the LSB method. The trick is to create three groups, namely regular (R), singular (S), and unusable (U) using the discriminant function (f) and the flip (F) operation, see Eq. (27).

$$I' \in R \Leftrightarrow f(F(I')) > f(I')$$

$$I' \in S \Leftrightarrow f(F(I')) < f(I')$$

$$I' \in U \Leftrightarrow f(F(I')) = f(I') \quad (27)$$

Based on Eq. (25), the flip operation is performed on the stego vector image. By using the $M = (0 \ 1 \ 1 \ 0)$ mask, we get R_M and S_M for positive masks, and R_{-M} and S_{-M} for negative masks. If the result is $R_M \approx R_{-M} > S_M \approx S_{-M}$ then it can be interpreted that it

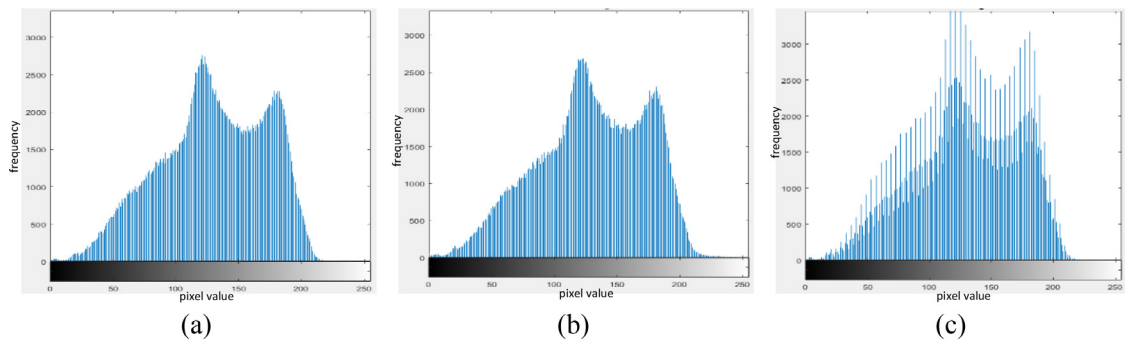


Fig. 4. (a) Original Baboon Image Histogram, (b) Stego Baboon Image Histogram using PVD [138], (c) Stego Baboon Image Histogram using LSB [138].

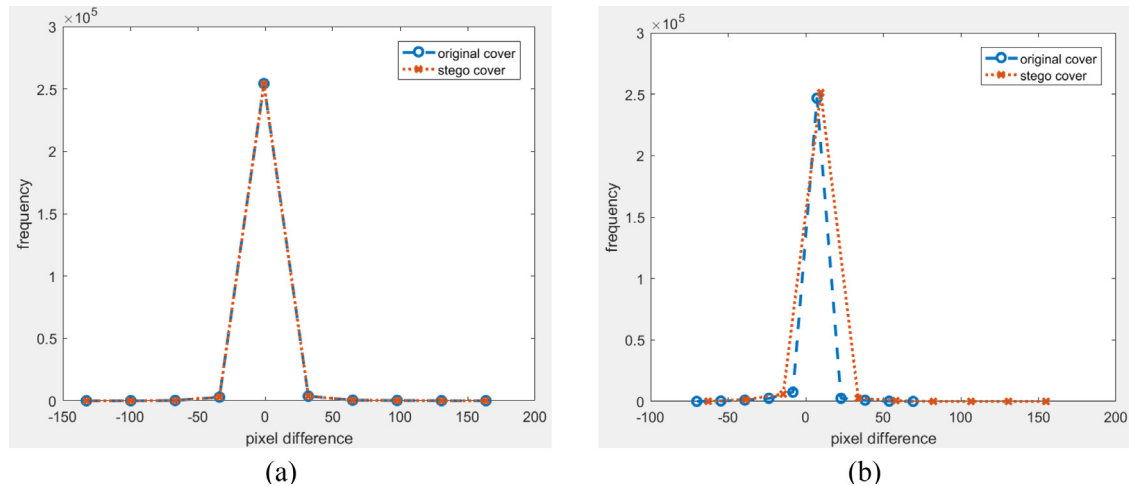


Fig. 5. Sample of PDH plot results (a) Undetected message using sophisticated PVD method (b) Detected message using standard PVD method.

is resistant to RS analysis. On the other hand, if the condition is $R_M - S_M > R_M - S_M$ exposed to RS analysis [38,39,54]. A plot of the RS value and embedding capacity is generally carried out to make it easier to read the steganographic method's safety quality results.

2.3.1.4. Pixel Difference Histogram (PDH) Analysis. The PDH method is one of the stochastic modulation-based steganalysis methods introduced in this research [142]. Standard steganography methods such as pixel value differencing (PVD) are generally not resistant to PDH analysis. This is because PDH compares the difference in the values of consecutive pairs of pixels. In grayscale images, the value of the pixel pairs is -255 to 255. In the original image and the stego image, each difference in the value of the pixel pairs will be calculated. Then a plot will be carried out, as shown in Fig. 5. If the two plots have subtle differences, then the message is not detected, but if there is a zigzag pattern or step effect on the plot, the message is detected [39,54].

2.3.1.5. Other Methods. There are several other specific steganalysis methods, but they are not popular enough to be used in security testing in steganography research. Some of them are sample pair analysis [143], weighted stego analysis [144], and structural steganalysis [145,146], used to detect the LSB method. Histogram characteristic function (HCF) for grayscale image [147], HCF-COM for colour image [148] to detect the LSB matching method. Method [149] to detect embedded messages using the stochastic modulation steganography method and many other methods.

2.3.2. Universal Steganalysis

Universal steganalysis methods are generally able to perform blind detection and are not focused on specific steganographic methods [140]. This is done by training and testing processes using classifiers on image features. This is quite similar to the classification method on images. It can be seen more clearly in Fig. 6. The classifier used can be a general classifier such as Fisher linear discriminant (FLD), support vector machine (SVM) [150], Neural network (NN) [140,151], or ensemble classifier (EC) [152] even using convolutional neural network (CNN) and deep learning [153]. Some of the widely used methods are presented as follows.

2.3.2.1. Subtractive Pixel Adjacency Matrix (SPAM). Image feature-based blind steganalysis was initially introduced in research [154]. Furthermore, it was developed into a more modern steganographic security measurement tool, namely SPAM, in research [155]. SPAM is designed to detect steganographic methods in the spatial domain, especially for the LSB matching method. There are two main stages of SPAM, the first using the first and second-order Markov chains to model the difference between adjacent pixels. Furthermore, SVM uses a subset of the transition probability matrices as a learning feature. Although this method is designed for detection in the spatial domain, from the test results, it can also detect domain transformations such as DCT [155]. The code to extract the SPAM feature can be downloaded at <http://dde.binghamton.edu/download/spam/>.

2.3.2.2. Textural Features based Universal Steganalysis. This steganalysis method is often referred to as Li-110D, published in [151]. Like SPAM, Li-110D is a universal steganalysis that performs detection based on texture features in grayscale images. In the Li-110D

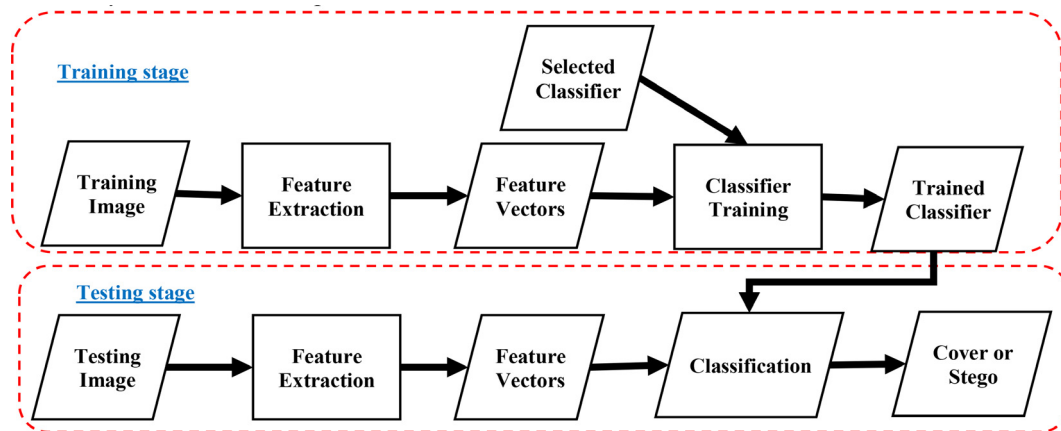


Fig. 6. Common stages of Universal Steganalysis.

feature, there are 110 features extracted from the typical histogram of an image's local linear transformation coefficient. These features are classified as FLD. This method can detect messages embedded with adaptive spatial domains or based on JPEG and JPEG2000.

2.3.2.3. Spatial-rich Model (SRM) Feature. This method was first proposed in research [156]. SRM is composed of 39 symmetrized sub-models quantized with three distinct quantization factors, totaling 34,671 features. Because of its high dimensionality, it is implemented using a machine learning method called an ensemble classifier [152], which is made up of multiple base learners. Here the ensemble classifier will be trained with two classes, cover and stego image. The model's correctness is assessed using the ensemble's unbiased estimate of the testing error, known as the 'out-of-bag' error (E_{OOB}). It is a reasonable approximation of the testing error. SRM feature extractor can be downloaded on the page http://dde.binghamton.edu/download/feature_extractors/. Furthermore, the SRM method was also developed into several methods such as: JPEG-Rich model (J+SRM), Cartesian-calibrated JRM (CC-JRM) [157], spatial-color rich model (SCRM) [158], steerable Gaussian rich model (SGRM) [159], Projection Spatial Rich Model (PSRM) [160], maxSRM [161], sigma-features [162], etc.

2.3.2.4. Universal JPEG Steganalysis Features. There are several steganalysis methods designed to detect the presence of messages stored in a stego image with a JPEG domain. One of the first steganalysis developed for this domain is CCPEV [163]. CCPEV is built from the merging Markov and DCT feature. Another steganalysis used in security testing of various steganographic research is the Contourlet-based steganalysis method (CBS) [164] extracted from DCT features and wavelets, DCT Residual (DCTR) [165] with 8000 feature dimensions, GFR [166] which is built from 2D Gabor filters for adaptive JPEG steganography, and the PHase-Aware pRojection Model (PHARM) [167] which is built based on the development of DCTR which consists of 12,600 feature dimensions. One of the newest methods is the maximum diversity cascade filter residual (MD-CFR), which combines features based on maximum diversity [168].

2.3.2.5. Deep Learning Method. Steganalysis research based on Convolutional Neural Networks (CNN) was developed in 2014 [153]. This research was initiated by Tan and Li and published in Tan and Li [169]. The results are not better than SRM but better than SPAM, even though this is the basis for developing CNN-based steganalysis methods. Furthermore, many methods of development, such as QianNet [170], XuNet [171], YeNet [172], Yedroudj-Net [173], ZhuNet [174], where ZhuNet managed to get bet-

ter performance from SRM. The steganalysis architecture is designed for the spatial domain. Furthermore, SRNet was also developed [175], which can work in the spatial and JPEG domains, where SRNet minimizes the use of heuristics for the steganographic noise extraction process. It can be concluded that the current deep learning-based steganalysis method has succeeded in surpassing the SRM+EC method [153]. DL-based steganalysis is still being developed today, such as GBRAS-Net [176], AG-Net [177], MixNet [178], and DRN [179], which is specifically for the spatial domain. Other methods that can work on both spatial and JPEG domains are OneHot [180], multiperspective progressive structure adaptation (MPSA) [181], and probability-guided network [182]. Besides being CNN-based, there is also a sophisticated GAN-based steganalysis method called NetS * [183].

2.3.3. Summary of Security Assessment

Steganalysis tools have significantly developed, from specifically designed to universally designed methods. Some measuring tools are also in the form of applications such as StegExpose. Table 4 presents the steganalysis method used in various steganographic research described in Table 6.

2.4. Robustness and its assessment

It is known that the steganography method has major challenges in imperceptibility, security, and payload. The robust aspect is actually only used as an added value. But in its development, in real cases, such as data transmission via social networks, lossy compression is carried out [119,134], or a network that has noise so that it distorts the cover image. This has become one of the open issues in the development of steganographic methods in the future. Logically, robust steganography is actually more challenging than robust watermarking is also taken into account [111]. Besides that, a blind extraction process with minimal and even no errors is also needed in steganography.

In contrast to watermarking which does not require 100% extraction results, and can use non-blind extraction. The meaning of robust in data hiding terminology refers to resistance to distortion attacks or manipulation of the cover, not security as described in section 2.3 or imperceptibility as described in section 2.1. Still, some studies use the word robust to refer to security and/or imperceptibility [112–114,190,191]. Although this is not completely wrong, it can confuse novice researchers, so it is better not to use the term robust when referring to security or imperceptibility. The robustness measurement is different from other aspects. If the other aspect measured is the stego image compared to the

Table 4

Map of the use of Security Assessment Tools in Image Steganography research in Recent Years.

Assessment Tools	Researches
RS Analysis	[3,31,34–36,38–41,43,47,53,54,58–61,63,77,82,83,93,98,100,103,109,113,116]
PDH Analysis	[3,31,34,35,39,43,47,54,60,61,63,65,82,83,100,109]
Histogram Analysis	[26,30,36,37,44,48,55–57,75,77,79,81–83,91,92,104,107,114,184,185]
Chi-Square Analysis	[30,34,36,44,48,56,66,97,113]
StegExpose	[36,91]
Other Specific Steganalysis	[30,32,34,36,41,48,56,61,68,75,83,91,95,98]
SPAM	[35,63,83,121,130,186]
Li-110D	[29,33,102,121]
SRM/J++SRM /maxSRM/ maxSRMd2	[83,111,119,121,124–126,133,136,183,185–188]
CCPEV	[110,111,115,129–131,135]
DCTR	[34,110,111,115,119–121,127,129,131,132,134,135,137,188]
SCRM/CRM	[117,123]
SGRM	[117,123]
GFR	[32,115,120,121,132,136,137,188]
DRN	[117]
SRnet	[30,95,134,189]
PHARM	[120,132]
Deep Learning Steganalysis (CRM, Ye Net, CMMF, CNN)	[89,183]
Other Steganalysis method	[32,111,124,125,136,137,187,189]

cover image, then the robustness measurement is carried out on the original and extracted messages. This indirectly also measures the quality of the extraction results. Regarding image watermarking, robustness tests generally use correlation or error extraction with and without attacks. But in the case of steganography, the extraction test is generally carried out without attacks, except for robust steganography, JPEG compression, or scaling can be performed because this method must be able to withstand noisy channels. The following is a measuring tool used to measure robustness in steganography.

2.4.1. Bit Error Ratio (BER)

BER is a measuring tool to calculate the bit error in the message. Because what is measured is a binary number, it can measure various forms of messages, such as text, images, audio, and even video. This is the simplest measuring tool, comparing the original message bits with the extracted message bits and then dividing by the original message bit length. The number of bits for the original and extracted messages should be the same. But if the extracted message bits are shorter or longer, then the remaining length is considered a bit error. BER can be calculated by adding the result XOR bit operation of the original message and the extracted message, see Eq. (28). The robustness value of the steganography method will be better if the BER value is close to 100%. In generic steganography, especially with the spatial domain, it should be able to extract 100% of the message if there is no distortion or manipulation of the stego image.

$$BER = \frac{\sum_{x=1}^{L_M} M_x \vee M'_x}{L_M} \times 100\% \quad (28)$$

Where L_M is total original message bits, x is message bits index, M is original message, M' is extracted message bits from stego image.

2.4.2. Character Error Ratio (CER)

CER is a measuring tool similar to BER, and only CER works every 8-bits or 1 byte. This is because what counts is the character error that occurs in the extracted message. This means that CER is specifically designed for error measurement of text-based messages. However, it can also be used to measure the difference in pixel values for each channel of image messages or other messages that are 8-bits in size. Just like BER, perfect CER must get 100% results.

2.4.3. 2D-Correlation Coefficient (CC)

CC is a robustness measurement tool that is often used to measure the correlation of messages, usually in the form of images using a median filter. In steganography, CC is not widely used unless the message is in the form of an image. CC is more widely used in image watermarking because, generally, watermarks are in the form of images. The way CC works do not measure based on the error that occurs but calculates the correlation of the original message image pixels with the extracted message. So that the CC value can be as long as it has a high correlation even though there are some errors. CC values range from -1 to 1, and a value of 1 is a perfect CC value. CC can be calculated by Eq. (29).

$$CC = \frac{\sum_x \sum_y (M_{xy} - \bar{M})(M'_{xy} - \bar{M}')}{\sqrt{(\sum_x \sum_y (M_{xy} - \bar{M})^2)(\sum_x \sum_y (M'_{xy} - \bar{M}')^2)}} \quad (29)$$

For the record, some research also uses the Normalized cross-correlation (NCC) measuring instrument. This measuring tool is slightly different because the size of the two images in NCC may not be the same. Where it is better that the image to be measured must be larger than the original image so that the normalization process is more meaningful. In this case, a more suitable measuring tool is the correlation coefficient.

2.4.4. Error Rate Extraction (R_e)

R_e is the most widely used measuring tool in testing message extraction in secure and robust steganography. This measuring tool serves to calculate the bit error that occurs in the extracted message. Basically, the formula R_e similar to BER, only not multiplied by 100%. If the error value is close to 0 it indicates that the error is getting smaller; if it is zero, then the extraction is done perfectly. can be calculated by Eq. (30).

$$R_e = \frac{N_e}{L_M} \quad (30)$$

Where N_e is the number of error message bits, L_M is the total message bits.

2.4.5. Other Robust Assessment

In addition to using the four measuring tools above, several studies use other tools to measure robustness, such as PSNR, MSE, SSIM, MSSIM, CF, and even StirMark Analysis. The measuring tool is generally used to measure messages in the form of images, so

Table 5

Map of the use of Robustness Assessment Tools in Image Steganography research in Recent Years.

Assessment Tools	Researches
BER	[35,37,38,46,54,57,89,93,116,117,122]
Error Rate Extraction(R_e)	[110,111,115,119,120,127–137,185,186]
CER	[46,94]
CC/NCC/NC	[32,88–90]
PSNR	[35,37,38]
SSIM	[37]
MSSIM	[89]
CF	[37]
StirMark Analysis	[56,75,91]

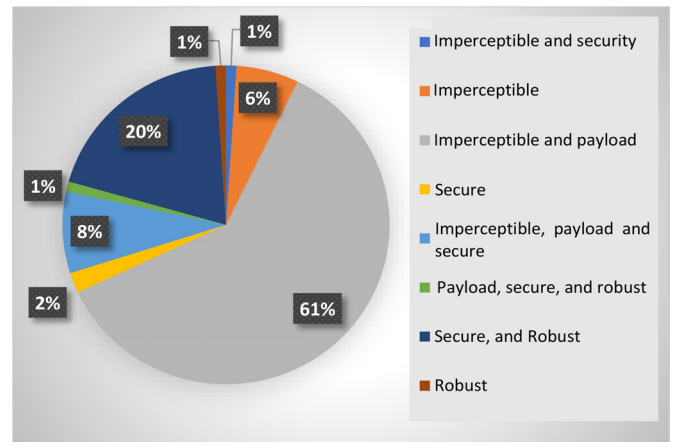
the quality of the extracted images is like measuring stego images. This additional measuring instrument actually does not really have a significant contribution, if the extraction results have been measured using R_e , BER or CER, because the purpose of steganography is to protect the message and ensure that the message reaches the recipient correctly.

Some things that are not quite right are when, in a paper PSNR, and MSE measurements are made on an attacked stego image, because the purpose of this measurement sometimes makes it unclear and confusing. Measurements on the original stego and the attacked stego image only aim to determine the level of damage to the attacked stego image. Measuring tools such as StirMark are also less precise when used in steganography because StirMark is designed for robust tests on watermarking. Table 5 summarizes the robustness measurement tools used in various studies.

3. Image Steganography Method and the Classification

First of all, in this section, Table 6 is presented, which summarizes state-of-the-art image steganography research published starting in 2015, which can be searched by the query (impercept* OR robust OR secure OR payload) AND Image AND (steganography OR "Data Hiding") on the socpus.com page. Based on the search results, there were many papers obtained, so several studies were selected with several exclusion criteria, namely 1) Studies without a clear contribution, measurement, and dataset (not just implementation or application), 2) Studies without a strong validation or including experimental results of image steganography (not watermarking), 3) Conference papers except conferences that specifically discuss data hiding or security, 4) Studies not written in English and not indexed by Scopus/WoS.

Table 6 contains much of the research reviewed and analyzed. These studies are a collection of research discussed in the previous section or will be addressed in the next section. We make all data into one table to show clear differences in each category of steganography research, which will be divided based on goals. In Table 6, it is clear that the development of steganography in the last five years varies greatly, not only in methods but also in measuring instruments. Based on assessment tools, the purpose of developing the steganographic method is also different. Some research also proposes interesting ideas by combining steganography and enhancement, zero embedding, or multiple stego output. However, some methods also indicate things that are not in accordance with the principles of steganography, such as semi-non-blind or non-blind extraction or color conversion that causes incomplete extraction. As a note, in Table 6, the result data presented on imperceptibility is limited to PSNR and SSIM only, while the payload is limited to BPP and PC. The rest is only the measuring instrument used. In addition, several papers with non-image steganographic titles, such as reversible data hiding, sometimes confuse some novice researchers. However, after we surveyed reversible data hiding (RDH) titles, generally has a fairly close pur-

**Fig. 7.** Goal Category of Image Steganography Method.

pose to steganography. The difference is only in the reversibility of the cover image, but indeed in its development, some of the other RDH methods have specific purposes and highlight other aspects not yet explained in this survey.

3.1. Based on Goal

In section 2, it has been explained that the steganographic measuring instrument is classified into four aspects to assess the quality of the steganographic method. However, not all steganographic studies use assessment tools from these four aspects, see Table 6. This proves that various steganographic methods were developed with different goals. In previous survey papers such as [4,11,20,28,194,195], It has been explained that steganographic images are classified by domain, retrieval behavior, adaptivity, cover dimension, cover format, secret format, method type (ML/DL/statistics), etc. However, the classification of steganographic images based on their goals has not been discussed explicitly. As discussed in section 2, steganography has four aspects: imperceptibility, payload capacity, security, and robustness. Based on the results of our review, there is a fact that payload capacity has a trade-off for each of the three aspects, especially imperceptibility. Not many of the proposed steganographic methods can improve the performance of these four aspects in a balanced way. Generally, the proposed method focuses more on one or two aspects. However, there may also be more than two aspects. Of the 97 studies reviewed in Table 6, we present them in a pie chart classified by goals in Fig. 7. This classification is especially done based on the research goals themselves and the assessment instruments used. The assessment instrument is the most important thing because this proves that the research objectives have been achieved. In Fig. 7, it appears that the two goals studied the most are imperceptibility and payload, and the second is secure and robust. The third category is quite complete and consists of three aspects, namely imperceptibility, payload, and security, and the other categories are not yet developed much. The proportions presented Fig. 7 does not mean that the largest portion (imperceptible and payload) is a more superior goal in steganography. But because the development is relatively simpler and relatively easier. Based on this data, we further discuss the top two categories in section 3.1.1. and 3.1.2.

3.1.1. Imperceptible and Payload Capacity Image Steganography

An increase in payload capacity generally must be accompanied by maintenance of imperceptibility or even an increase in imperceptibility. This method is relatively more popular and has been widely developed because it is relatively easier and can be

Table 6
Recent Image Steganography Method.

Ref	Year	Method	D	R	Assessment Tools				Datasets	Goals	Notes
					Imperceptibility	Payload	Security	Robustness			
[25]	2017	LSB based + Edge	S	N	PSNR up to 40.16 dB for 2.79 BPP	BPP up to 4.11	N/A	N/A	USC-SIPI / Standard Image test	I+P	-
[26]	2020	LSB based + Hamming Code	S	N	PSNR over 60dB	PC up to 30224 bits	N/A	N/A	UCID-Image Database	I+P	-
[27]	2018	LSB + LoG	S	N	PSNR up to 62.17 dB for 0.08 BPP	BPP up to 3.10	N/A	N/A	USC-SIPI / Standard Image test	I+P	-
[29]	2016	IWT or LSB + Edge + XOR Function	S/T	N	PSNR more than 52 dB in N-BPP IWT domain and 50 % embedded, MSE, WPSNR	BPP and Percentage Embedding	Li-110D	N/A	USC-SIPI / Standard Image test + BOWS2 database	I+P	-
[40]	2018	LSB + PVD with compression and encryption	S	N	Avg PSNR 36.38dB for 1049240bits capacity, SSIM> 0.94 for 4 BPP PSNR $\approx$ 44dB,	Up to 4BPP	RS Analysis	N/A	USC-SIPI / Standard Image test	I+P	-
[51]	2018	Histogram Shifting and Pixel Mapping	S	Y		BPP $\approx$ 1.5 PC 144,150- 146,720 bits	N/A	N/A	USC-SIPI / Standard Image test	I+P	-
[64]	2021	LSB substitution	S	N	PSNR $\approx$ 43dB for $\approx$ 2 BPP, MSE, SSIM	BPP 0.31-5.20	N/A	N/A	USC-SIPI / Standard Image test	I+P	-
[65]	2018	PVD based	S	N	PSNR 44.21- 53.99dB	Avg PC $\approx$ 400,000 bits per layer (for 8-bits)	PDH Analysis	N/A	USC-SIPI / Standard Image test	I+P	-
[66]	2021	LSB + chaotic map	S		PSNR > 50dB KL Div, CQ, SC, IF, AD	BPP 0.67	Chi-square	Salt and Pepper	USC-SIPI / Standard Image test	I	-
[67]	2016	Modified LSB + Edge	S	N	PSNR 39.0025-52.8913dB	BPP 1.1164-3.1769, PC 73,163-208,200 bits	N/A	N/A	USC-SIPI / Standard Image test + Seam Carving Original Q75	I+P	-
[68]	2019	Linear Time Digit based	S	N	Avg PSNR 38.74 dB for 4BPP, UIQI, MAE	BPP 4, PC	Weighted Stego-Image (WS) Analysis	N/A	USC-SIPI / Standard Image test	I+P	-
[30]	2021	LSB + Edge based DL	S	N	PSNR up to 37.11 dB, SSIM up to 0.95 in 3.45 BPP	BPP >3.15	RS, Primary sets, Chi-square, Sample pair, Histogram, SRnet	N/A	USC-SIPI + Berkeley Segmentation Dataset and Benchmark (BSDS 500)	I+P +S	-
[31]	2017	Adaptive PVD	S	N	Avg PSNR 50.93dB and UIQI 0.999 for 1.83BPP	Avg BPP 1.83, avg PC 1445645 bits	RS, PDH	N/A	USC-SIPI / Standard Image test	I+P	-
[32]	2019	Framelet transform and bidiagonal SVD	T	N	PSNR > 69dB, SSIM = 0.9999, embedding in high coefficient and SV MSSIM, IF, UIQI, FSIM	256 Kb	Wavelet-Based Steganalysis (WBS) [150], Contourlet Based Steganalysis (CBS), an optimal linear predictor of coefficient magnitude, GFR	Tested on various attacks, but cannot get NC = 1 without attack NC, BER, and PSNR to measure stego image damage	USC-SIPI/ Standard Image test Washington image database	S+R	Cannot extract perfectly, indicated use non-blind extraction
[33]	2016	LSBM + STC + Edge	S	N	PSNR $\approx$ 50-68dB, MSE, wPSNR	PC 3251-32600 bits, Embedding Rate 5-50%	Li-110D	N/A	Medical (MRI) gray	I+P +S	-
[34]	2018	WPD + Edge + LSB	T+S	N	PSNR $\approx$ 31-48dB	BPP $\approx$ 1-3.3, PC up to 84,902 bytes	RS, PDH, DCTR with 0.1-0.2 BPP	N/A	USC-SIPI / Standard Image test + BossBase	I+P +S	-
[35]	2021	PVD + MF	S	N	Avg PSNR around 39 dB, Avg SSIM 0.98	Avg BPP 3.10 Avg PC more than 800000bits tested up to 4BPP	RS, PDH, SPAM,	BER and PSNR for salt and pepper attack	USC-SIPI / Standard Image test	I+P +S	-
[36]	2021	Catalan transform	T	N	PSNR $\approx$ 48.6dB, SSIM=0.99 for 1 BPP, MSE, NCC		StegExpose tool, histogram analysis, Primary Sets, Sample pair analysis, RS analysis, Chi-square attack	N/A	USC-SIPI / Standard Image test + RGB-BMP Steganalysis	I+P	-
[37]	2020	DT-CWT + SFTA + GLCM + KNN	T	N	Avg PSNR more than 51dB, Avg SSIM more than 0.9993 MSE, CF	BPP 7.875	Histogram	PSNR, SSIM, CF, BER *image message *Avg PSNR $\approx$ 38dB $\rightarrow$ not perfect extraction and need position vector	USC-SIPI / Standard Image test + Seam Carving Original Q75	I+P	Indicated use semi non-blind extraction

(continued on next page)

Table 6 (continued)

Ref	Year	Method	D	R	Assessment Tools				Datasets	Goals	Notes
					Imperceptibility	Payload	Security	Robustness			
[38]	2019	n-Rightmost Bit Replacement Image	S	N	Avg PSNR 34.83-51.21 Avg SSIM 0.99-0.97	BPP 1.0-4.0 PC 262,144 - 1,048,576 bits	RS	BER and PSNR for Salt and pepper attack	USC-SIPI / Standard Image test	I+P	-
[39]	2019	Improved LSB Matching	S	Y	Avg PSNR 49.44-51.30 Avg SSIM 0.9977-0.9988	PC 524,288 bits for 512×512 cover image	RS, PDH	N/A	USC-SIPI / Standard Image test	I+P	Output dual stego image
[41]	2019	LSB + Edge with Threshold	S	N	PSNR ≈47dB, SSIM ≈0.98 MSE, KL Div, Cross-correlation coefficient, Kurtosis, Standard Deviation, Mean, entropy	BPP 2.0	RS, Sample Pair Analysis	N/A	USC-SIPI / Standard Image test + Corel 10 k Dataset	I+P	-
[42]	2020	IWT-SVD and DWT-SVD	T	N	PSNR 61-71 dB, SSIM 0.89-0.99, BRISQUE, MSE	claimed 25.42 BPP, but embedding only on singular of LL sub-band	N/A	N/A	Fractal Image + USC-SIPI / Standard Image test	I+P	Indicated use non-blind extraction
[43]	2019	HOG and PVD-LSB	S	N	PSNR >34dB, UIQI > 0.799 for gray image PSNR >42dB, UIQI > 0.99 for color image	PC 100,000 bytes	RS, PDH	N/A	USC-SIPI / Standard Image test	I+P	-
[44]	2020	LSB + 1D chaotic map	S	N	PSNR >35dB, UIQI, IF, MSE, Correlation, Homogeneity, Entropy, Contrast, Energy	Max PC 50,000 bytes	Chi-Square, Histogram	N/A	USC-SIPI / Standard Image test	I+P	-
[45]	2018	Hamming Code + Edge + LSB	S	N	PSNR >45 dB	BPP≈1.2	N/A	N/A	USC-SIPI / Standard Image test	I+P	-
[46]	2022	LSB + Hybrid Edge Dilation	S	N	Avg PSNR ≈ 39dB, Avg SSIM ≈ 0.999	Avg BPP=1.4531-2.4531, PC= 95,231- 160,767 bits	N/A	CER, BER without attack/ extraction measurement	USC-SIPI / Standard Image test	I+P	-
[47]	2021	Retracing Extended-Sudoku	S	N	Avg PSNR 39.63, SSIM 0.9908 for 3.16 BPP, UIQI, MSE	Up to 3.1699 BPP, PC 830,976 bits	RS, PDH	N/A	USC-SIPI / Standard Image test	I+P	-
[48]	2022	LSB + Distinction Grade Value (DGV)	S	N	PSNR > 69dB, SSIM 0.99, NCC	PC 49,152 bytes	Chi-square, Histogram, bit plane	N/A	USC-SIPI / Standard Image test	I+P	-
[49]	2018	Bit differencing + LSB matching	S	N	Avg PSNR 44.07dB, MSE, UIQI	BPP 2, PC 524288 bits	N/A	N/A	USC-SIPI / Standard Image test	I+P	-
[50]	2019	DE+MF	S	Y	PSNR ≈ 53-56 dB	BPP ≈ 0.06 - 0.14	N/A	N/A	USC-SIPI / Standard Image test	I	-
[52]	2019	AMBTC + Hamming Distance + PVD	S	N	PSNR ≈ 23-34 dB	PC 140,130 - 261,918 bits	N/A	N/A	USC-SIPI / Standard Image test	I+P	-
[53]	2019	PVD + MF	S	N	Type 1, PSNR 42.04, SSIM 0.99 (Avg) Type 2, PSNR 38.36, SSIM 0.98 (Avg)	Type 1, BPP 2.2, PC 577,832 bits (Avg) Type 2, BPP 3.1, PC 823,544 (Avg)	RS	N/A	USC-SIPI / Standard Image test + CVonline	I+P	-
[54]	2022	LSB Matching + Pixel Difference	S	Y	Type 1 PSNR > 51dB, SSIM >0.99 (Avg) Type 2 PSNR > 37dB, SSIM > 0.96 (Avg)	Type 1, PC 524,288 bits Type 2, avg PC up to 1,580,375 bits	RS, PDH	BER of salt and pepper attack	USC-SIPI / Standard Image test	I+P	output: a dual stego image
[55]	2020	Bi-Orthogonal Wavelet Transform + GA	T	N	PSNR > 50dB, MSE, PCC, VIF, Standard Deviation	Not explained, but based on our calculations it is 0.5BPP	Histogram	N/A	USC-SIPI / Standard Image test	I+P	-
[56]	2019	Multi-bit modulo operation	S	N	PSNR 34.7dB for 4BPP, LMSE, NCC	1048576 bits	Histogram, Bit plane, Chi-square, Sample Pair, Weighted Stego, TriplesAnalysis	Stirmark	USC-SIPI / Standard Image test	I+P	-
[57]	2021	LSB with modulus operation +BES+BRISK	S	N	PSNR>90dB, SSIM >0.99	Average capacity for standard image 569 bits	N/A	BER without attack, and with rotation, scaling, adding noise, and JPEG compression attack	L3IDocCopies + DSSE 200 + LRDE DBD + LRDE DBD + USC-SIPI / Standard Image test	I	-
[58]	2022	LSB-MR + EMD	S	Y	Avg PSNR 51.14 dB, avg SSIM 0.9975	BPP 0.9997, PC 524,267 bits	RS	N/A	ALASKA v2	I+P	output: a dual stego image

(continued on next page)

Table 6 (continued)

Ref	Year	Method	D	R	Assessment Tools				Datasets	Goals	Notes
					Imperceptibility	Payload	Security	Robustness			
[59]	2019	LSB + PSO	S	N	Avg PSNR 56.60dB	BPP 4.0	RS	N/A	USC-SIPI / Standard Image test	I+P	-
[60]	2022	Block based 3×3 LBP-RDH	S	Y	Avg PSNR 52.36dB, Avg SSIM 0.9835	BPP 1.98	RS, PDH	N/A	Random Grayscale Image	I+P	output: a dual stego image
[61]	2021	LSB based Shadow Image and XOR feature	S	Y	Avg PSNR >52dB, Avg SSIM > 0.997	PC 786,432 bits	RS, PDH, Bit Pair Analysis	N/A	USC-SIPI / Standard Image test	I+P	output: a triple stego image
[63]	2022	LSB based XOR features	S	N	Avg PSNR 51.88dB, Avg SSIM 0.99, UIQI	PC 262,144 bits	RS, PDH, SPAM	N/A	USC-SIPI / Standard Image test	I+P	-
[74]	2018	LSB + bit Flip	S	N	PSNR > 43dB, UIQI > 0.9 for 2 BPP	BPP 2, PC 524,288 bits	N/A	N/A	USC-SIPI / Standard Image test	I+P	-
[75]	2020	PVD based	S	N	Avg PSNR 41.59dB, SSIM 0.992, MSE, BER, UIQI, NCC	Avg BPP 2.94	Histogram, Color Frequency, StegExpose	StirMark	USC-SIPI / Standard Image test	I+P	-
[77]	2018	LSB + Edge	S	N	Avg PSNR≈44dB, SSIM≈0.998, MSE, FSIM	BPP 1.25	N/A	N/A	USC-SIPI / Standard Image test	I	-
[81]	2021	LSB + bit Flip	S	N	PSNR > 33dB, SSIM, MSE, KL Div	PC 1,048,576bits	Histogram Analysis	N/A	USC-SIPI / Standard Image test	I+P	-
[82]	2015	PVD+MF	S	N	WPSNR, KL Div	PC 420,577 kbits	Histogram, RS	N/A	USC-SIPI / Standard Image test	I+P	-
[83]	2022	IWT/DWT + Genetic Algorithm (GA)	T	N	PSNR >49dB for 0.75 BPP, SNR, KL Div, SC, SNR, VIF	BPP 0.75	RS, PDH, SPAM, SRM, maxSRM, maxSRMd2	N/A	USC-SIPI/ Standard Image test, NRCS dataset	I+P +S	-
[85]	2018	Histogram Shifting + contrast enhancement	S	Y	PSNR=20.76–25.54dB, SSIM=0.853–0.936, BRISQUE, C-DIIVINE, NFERM	BPP up to 0.856	N/A	N/A	USC-SIPI + Kodak Lossless True Color Image Suite	I+P	Combined with image enhancement
[87]	2018	Adaptive LSB, RGB→HSV→RGB	S	N	Avg PSNR 42.455, SSIM ≈0.99, MSE, NCC, MOS	PC 6,144 bytes	N/A	N/A	USC-SIPI / Standard Image test	I+P	Color conversion may affect imperfect extraction
[88]	2018	Finite Ridgelet Transform –DWT	T	N	PSNR > 58dB for 8BPP with IF 0.2	Claimed 8 BPP	N/A	NC with many attacks like watermarking robustness testing	Standard image dataset + USC-SIPI	I+P	indicated using non-blind extraction
[89]	2021	Redundant IWT + Laplacian pyramid + histogram shifting	T	N	PSNR > 49dB for 1BPP	Up to 4BPP	CRM, YeNet and CMMF	NCC, BER, MSSIM dengan berbagai macam attack seperti watermarking	Standard image dataset + USC-SIPI	P+S +R	Extraction test results without attack are not explained
[90]	2022	DCT	T	N	PSNR > 71 dB and SSIM > 0.998 for 1024 bits, NCC	Up to 8192 bits	N/A	NCC =1 for JPEG compression Q10-Q90 with payload 1024 bits	Standard image dataset + USC-SIPI + BSDS300	R	-
[91]	2021	DE + Interpolation	S	Y	PSNR 29.1141 dB to 36.2917 dB, SSIM 0.9726 to 0.9974, UIQI, NCC	BPP 1.9375 –3.2640	Histogram, StegExpose	StirMark Benchmark 4.0 analysis	USC-SIPI / Standard Image test	I+P	-
[92]	2022	ASCII code matching	S	Y	Claimed without Distortion, PSNR ∞, SSIM 1, MSE 0	Claimed unlimited capacity	Histogram	N/A	USC-SIPI + private dataset	I+P	Zero / without embedding
[93]	2022	Inverted LSB + Edge Pattern	S	N	Avg PSNR ≈ 56dB, SSIM ≈ 0.999, MSE	BPP 1	RS	BER without attack/ extraction measurement	USC-SIPI / Standard Image test	I+P	-
[94]	2019	LSB + Edge Dilation	S	N	Avg PSNR 58.957dB, Avg SSIM 0.9997, MSE	Avg BPP 1.41954	N/A	CER without attack/ extraction measurement	USC-SIPI / Standard Image test	I+P	-
[95]	2021	LSBMR	S	N	Avg PSNR 61.398 dB (various payload), SSIM >0.99 (max Payload)	PCC >7×10 ⁴	SRNet	N/A	USC-SIPI / Standard Image test + BOSSbase 1.01	I+P +S	-
[96]	2020	Enhanced LSB (with lossless compression)	S	N	PSNR > 50dB, MSE, RMSE	N/A, just an embedding bit requirements	N/A	N/A	USC-SIPI + Private	I+P	-

(continued on next page)

Table 6 (continued)

Ref	Year	Method	D	R	Assessment Tools				Datasets	Goals	Notes
					Imperceptibility	Payload	Security	Robustness			
[97]	2019	EMD	S	N	Avg PSNR 55.71 dB SSIM > 0.9	BPP 1.59	Chi-square	N/A	USC-SIPI / Standard Image test	I+P	-
[98]	2017	LSB + Edge	S	N	PSNR > 50 dB	BPP≈1	Visual attack, RS, high order statistical analysis, Farid 72D	N/A	USC-SIPI / Standard Image test	I+P	Use semi-non-blind extraction
[99]	2017	Adaptive LSB	S	N	Avg PSNR 45.031dB, Avg SSIM 0.955, RMSE, MSE, NCC	PC 8 KB	N/A	Salt and pepper	USC-SIPI / Standard Image test	I+P	-
[100]	2018	PVD based	S	N	Avg PSNR type 1 = 39.55dB, type 2 = 37.22dB, UIQI	Avg BPP type 1 = 3.0, type 2 = 3.31, PC type 1 =2,361,368 bits, type 2 =2,603,604 bits	RS, PDH	N/A	USC-SIPI / Standard Image test	I+P	-
[101]	2017	Image Interpolation + LSB + OPAP	S	Y	PSNR = 45.77 dB (reversible), 45.26 dB (irreversible) * *on Lena's Image	PC 155,598 bits (reversible), 393,660 bits (irreversible) * *on Lena's image (not Max)	N/A	N/A	USC-SIPI / Standard Image test	I+P	-
[102]	2019	LSB+Edge	S	N	PSNR > 48dB, SSIM > 0.99, MSE	PC > 183500 bits	Li-110D	N/A	BOWS2 database	I+P +S	-
[103]	2018	LSB + Location Map	S	Y	PSNR ≈ 51dB for One-layer embedding, MSSIM	BPP ≈ 1.9 One-layer embedding	RS	N/A	USC-SIPI / Standard Image test	I+P	-
[104]	2022	IWT	T	N	Avg PSNR 59.4653, Avg SSIM 0.85273, MSE, RMSE	PC 10,000 char/bytes	Histogram	N/A	DICOM Medical Images	I+P	-
[105]	2021	LSB + GA	S	N	PSNR > 78dB, SSIM > 0.99	BPP 1	N/A	N/A	USC-SIPI / Standard Image test	I	-
[106]	2018	LSB + Hybrid Edge	S	N	PSNR = 33.9699dB	BPP 3.13	N/A	N/A	USC-SIPI / Standard Image test	I+P	-
[107]	2017	Overlapped PVD	S	N	Avg PSNR 32.79dB	Avg PC > 1,990,527 bits	Histogram	N/A	USC-SIPI / Standard Image test	I+P	-
[108]	2018	LSB + cuckoo search + chaotic map	S	N	Avg PSNR 33.66, SSIM 0.90 for 90% Payload	100% Payload 4 BPP	N/A	N/A	USC-SIPI / Standard Image test	I+P	-
[109]	2017	Parity-bit PVD	S	N	Avg PSNR 38.1dB	BPP 3.0	RS, PDH	N/A	USC-SIPI / Standard Image test	I+P	-
[110]	2021	DCT + RS encoding + STC	T	N	PSNR > 45 dB	PC 787,538 bits 0.001 – 0.10 bpnzAC	DCTR, CCPEV features	Error rate from JPEG compression Q65, Q70, Q75, Q80, Q85	BossBase + UCID + IS-tego100K	S+R	-
[111]	2019	DCT+STC+ECC+ Huffman	T	N	PSNR > 42 dB, SSIM > 0.997	0.001 – 0.10 bpnzAC	CCJRM, CCPEV features	Error rate from JPEG compression Q65, Q75, Q85	BossBase	S+R	-
[115]	2019	J-UNIWARD+UERD+ STC+ Coefficient adjustment	T	N	PSNR > 35 dB for 0.5 bpnzAC	0.05 – 0.5 bpnzAC	DCTR, GFR, CCPEV	Error rate from JPEG compression Q75, Q95	BossBase + UCID	S+R	-
[116]	2022	PVD +MF + Edge Pattern	S	N	Avg PSNR 40.28 dB, SSIM 0.98	BPP 3.18	RS	BER = 0 without attack	USC-SIPI / Standard Image test	I+P	-
[117]	2020	Dual-Tree Complex Wavelet Transform based super-pixel and coefficient-plane-selection	T	N	PSNR>50 dB, SSIM>0.92, CF, UIQI, EPI	Claimed up to 31.329BPP	CRM, SGRM	BER of Rotation, Scale, JPEG compression attack	Dataset from [192,193]	I+P +S	Indicated use non-blind extraction and not perfect extraction

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Table 6 (continued)

Ref	Year	Method	D	R	Assessment Tools				Datasets	Goals	Notes
					Imperceptibility	Payload	Security	Robustness			
[118]	2022	Scalable DE	S	Y	PSNR 45.6-60.3dB (mid 52.9dB)	PC 20.2k-110.1kbits (mid 65.25kbits)	N/A	N/A	USC-SIPI/ Standard Image test	I+P	-
[119]	2021	Compression resistant principle based adaptive steganography (CRPAS) based on ECC+STC	T / S	N	PSNR > 50dB for 0.01 bpnzAC	0.01-0.1 bpnzAC and BPP	DCTR, SRM	Error rate from JPEG compression Q65, Q75, Q85, Scaling 75%, 87.5%, 200%, 400% Gaussian noise 0.001, 0.005, 0.01, 0.03 Error rate from JPEG Q75, JPEG Q85, JPEG Q90	BossBase and UCID	S+R	-
[120]	2021	AutoEncoder with Adaptive BCH Encoding	T	N	PSNR > 52 dB for 0.3 bpnzAC)	0.05-0.3 bpnzAC	DCTR, GFR, PHARM		BossBase	S+R	-
[121]	2017	EMD based + critical functions	S	N	Avg PSNR >55dB, avg SSIM >0.998 for 0.5BPP	BPP 0.5	Li-110D, SPAM, SRM	N/A	USC-SIPI/ Standard Image test + UCID + McGill + Ground Truth	I + S	-
[122]	2019	Quadtree PVO	S	Y	PSNR > 60dB SSIM=0.9979 for Lena image with payload 30000 bits	Around 30000bits based on the image histogram	N/A	BER of salt and pepper attack, Gaussian and speckle	USC-SIPI/ Standard Image test	I	-
[123]	2020	Amplifying Channel Modification Probabilities (ACMP)	S	N	N/A	0.1-0.5 BPP	SCRM, SGRM	N/A	BossBase	S	-
[126]	2016	MiPOD	S	N	N/A	0.05-0.5 BPP	SRM, MaxSRMd2	N/A	BossBase	S	-
[133]	2018	Cost Function in J-UNIWARD	T	N	N/A	0.1 – 0.5 bpnzAC	JSRM	Error rate from JPEG compression Q75, Q85, Q95	Brain MRI Images, BossBase	S+R	-
[134]	2022	MINImizing Channel Error Rate (MINICER)	T	N	N/A	0.1 – 0.5 bpnzAC	DCTR, SRnet	Error rate from JPEG recompression Q70→Q70	BossBase	S+R	-
[135]	2020	ECC+STC based on DCT	T	N	N/A	0.01-0.1 bpnzAC	DCTR, CCPEV features	Error rate from JPEG compression Q65, Q75, Q85,	BossBase	S+R	-
[136]	2018	Generalized Uniform Embedding Distortion (GUED)	T	N	N/A	0.1-0.5 bpnzAC	GFR, CC-JRM, J+SRM, histogram	Error rate from JPEG compression Q75, Q95	BossBase	S+R	-
[137]	2018	Block Entropy Transformation (BET)	S + T	N	N/A	0.1-0.5 bpnzAC	GFR, SCA-GFR, DCTR, CC-JRM	Error rate from JPEG compression Q75, Q95	BossBase	S+R	-
[128]	2018	Compression-resistant Domain Constructing + RS-STC Codes	T	N	N/A	0.01-0.1 bpnzAC	N/A	Error rate from JPEG compression Q65, Q75, Q85	BossBase	S+R	-
[129]	2020	Generalized dither Modulation-based robust Adaptive Steganography (GMAS)	T	N	N/A	0.01-0.5 bpnzAC	DCTR, CCPEV	Error rate from JPEG compression Q65- Q85	BossBase	S+R	-
[130]	2018	DCT+RS coding + STC	T	N	N/A	0.01-0.1 bpnzAC	SPAM, CCPEV	Error rate from JPEG compression Q65, Q75, Q85	BossBase	S+R	-
[131]	2016	DCT+EC Coding+STC	T	N	N/A	0.01-0.1 bpnzAC	CCPEV, DCTR	Error rate from JPEG compression Q65, Q75, Q85	BossBase	S+R	-
[132]	2019	JPEG Compression ResIstant Solution (with Bch code)	T	N	N/A	0.05-0.3 bpnzAC	DCTR, GFR, PHARM	Error rate from JPEG compression Q75, Q85, Q90	BossBase	S+R	-
[185]	2015	Uniform Embedding Revisited Distortion (UERD)	T	N	N/A	0.1-0.5 bpnzAC	J+SRM, histogram	Error rate from JPEG compression Q75, JPEG Q95	BossBase	S+R	-
[186]	2019	Zernike Moment + Dither Modulation	S	N	N/A	128 bits	SPAM, MaxSRM	Error rate from scaling attack	BossBase	S+R	-
[188]	2020	Steganographic Polar Codes (SPC)	S+T	N	N/A	0.1-0.5 BPP	SRM, DCTR, GFR	Error rate from JPEG compression Q75	BossBase	S+R	-

tested only with standard test images. The method with this aim is mostly using the spatial domain method by highlighting the imperceptibility and payload capacity measurement tools. It can be seen in Table 6 that embedding 4BPP or more on average it has not been able to produce PSNR reaching 35dB and SSIM 0.9, but if at least it reaches BPP = 3 to 3.5, as in the study [35,53,109,116]. One of the steganographic methods that has a huge capacity and good imperceptibility is research [54]. The average embedding capacity is up to 1580375 bits on a 512×512 grayscale host image with PSNR values of 37dB and SSIM 0.96. This capacity is more than 6BPP, but this method produces four stego images. This idea was actually developed from research [39], where a dual stego image is generated with one cover to increase the capacity and imperceptibility of reversible data hiding (RDH). RDH is similar to steganography, but RDH can recover the cover image and secret data completely. Due to these recovery conditions, the complexity of RDH method is generally more complicated, and its capacity is relatively smaller than generic steganography. In Table 6, the steganographic methods marked with a Y label on the reversibility field can be grouped into RDH. Another note is that the research in Table 6 is marked with the label I+P+S on the field goal, as in research [30,33–35,83,95,117]. This means that the proposed method has excellent imperceptibility and capacity, has been tested with several universal steganalysis tools, and is proven to be quite secure. Why does this need attention? Because most research with the goal of I + P only performs security tests with specific standard steganalysis, even some research does not perform security tests with steganalysis.

Steganography and watermarking have many similarities, and both are data hiding methods. This sometimes makes researchers less sensitive in distinguishing how the two work. According to the theory, steganography should be extracted blindly. Besides that, the message should also be able to be fully retrieved [2]. Of the several methods presented in Table 6, some methods perform color conversion, embed part of the message (such as singular or certain subbands), or use certain keys to perform an extraction. Some also do not explain the details of the extraction process. So we have speculation that this method is unsuitable if it is called steganography because it uses a non-blind or semi-non-blind extraction method, and the message may not be fully retrieved, as in research [37,42,57,88,117,184]. In these methods, the majority also use images as messages, because messages in the form of images are more tolerant of differences in results that visually look the same. Another impressive thing is the claim of super large capacity like research [37] = 7.875 BPP, [42] = 25.42BPP, dan [117] = 31.329BPP. Logically, large distortion will occur in the image because one pixel contains eight bits for each layer. If it has a capacity of more than eight for each layer, it is certainly impossible to do blind extraction. There is also a steganographic method without distortion [92]. This method does not do embedding but creates a kind of “encryption” with certain operations based on the host image and secret message to get the extraction key.

Research with the title of robust image steganography such as [56], we group it in this class because the payload generated is huge with a relatively excellent PSNR, and the word robust in the title refers to secure, where the level of security is tested on a specific steganalysis measuring tool on LSB. Another note that needs to be considered is the idea of increasing the capacity of steganography by performing lossless compression, such as [96]. This will definitely increase the PSNR value because compression will automatically minimize the capacity so that the distortion will be reduced. Actually, from the side of the steganography method, this does not have a major contribution because it only implements the compression method in steganography. There have not been many security tests using universal steganalysis, and the latest steganalysis in research on steganography which tends to favor impercep-

tibility and high payload, this makes it an opportunity to make a significant contribution to this field.

3.1.2. Secure and Robust Steganography

The steganographic method is classified as secure if tested with various universal steganalysis tools. This is in accordance with the theory and principle that the steganographic method must have undetectability abilities [1,2]. So it is not appropriate if steganographic research claims the method is secure because it is only combined with cryptographic/encryption methods, for example, in research [40,114,196–202]. As a note, in research [114], the term robust is used, but it means secure. When the steganography method is combined with cryptography, it does not directly mean that a steganographic method is said to be secure. Although logically, this is a two-layered protection, combining existing cryptography and steganography methods and even compression combined does not have a significant scientific contribution unless there is a novelty in each method. The meaning of secure in steganography and cryptography is different. In secure steganography, it means the undetectability ability of steganalysis attacks. This can be tested with the measuring tools described in sub-chapter 2.3. Meanwhile, security measures in cryptography (especially on images) include entropy analysis, NPCR, UACI, avalanche effect, correlation coefficient, keyspace, and key sensitivity analysis, etc.

Robust steganography is the closest steganography to watermarking. The similarity is the robustness of the message when image manipulation occurs. Robustness in watermarking is designed to resist various kinds of attacks/image manipulation. The effect is that the copyright embedding process in watermarking can give a more significant distortion effect than steganography. Robustness in steganography is generally focused on compression and scaling/resize attacks. This is done so that the undetectability of steganography can also be maintained. Robust steganographic methods are generally also designed to resist various universal steganalysis.

In Table 6, a secure steganography method has been presented, marked with an 'S' label, and robust steganography, marked with an 'R' label on the field goals. If it is seen that almost all robust steganography methods are also secure, the difference is in robust steganography using transform domains, especially JPEG. These studies generally do not measure imperceptibility, only in some studies such as [32,90,110,111,119,120]. Although using the term robust, it will not be suitable in steganography if robustness testing is carried out with various kinds of distortion as in research [32,89]. And it will be less in accordance with the principles of steganography when all steganographic methods are indicated to use non-blind extraction, even though it is not explicitly explained in the research.

Most of the secure steganography research uses the BossBase dataset. There are other datasets such as UCID, IStego 100K, and sometimes combined with standard image tests. This is because testing on secure steganography generally uses universal steganalysis, a learning process requiring large datasets. So it is not enough to only use a limited standard test image dataset. Some of the universal steganalysis widely used in secure steganography are SRM, JSRM, DCTR, GFR, CCPEV, PHARM, and so on. Meanwhile, in secure and robust steganography, a robustness test is added by measuring the extraction error after the stego image is compressed by JPEG, generally from Q65 to Q95.

Several methods that are widely used as references for the development of secure and/or robust steganography are HUGO [203], wavelet obtained weights (WOW) [204], universal wavelet relative distortion (UNIWARD) [205] developed in the spatial domain (S-UNIWARD) and JPEG domain (J-UNIWARD), HILL [206], dst. The main mission of these methods is to assign low costs to coefficients in complex areas and high costs to coefficients in clean ar-

eas of the image, also known as the “Complexity First Rule” [207]. But the mission of methods based on distortion function frameworks, especially STC based, only works well on lossless channels. This operation will fail STC message extraction because the stego image is modified during transmission [208]. Many steganographic methods have been developed from these methods, which are designed to be more robust and secure when transmitted on lossy channels, especially on social media networks such as in this study [110,119,120,134,135]. But until now, getting perfect extraction accuracy after compression is still an open issue. Especially if you look more closely at Table 6, the average robust and secure steganographic has a much smaller payload. This is a very open challenge in the future to achieve robust, secure, high payload and imperceptible steganography.

3.2. Based on Domain

The difference in the domain of the steganographic method affects the performance and purpose of steganography. In Table 6, field (D) shows the main domain of the proposed steganographic method. There are two domains: spatial (S) and transform (T). In more detail, the discussion of each domain is described below.

3.2.1. Spatial Domain

The spatial domain is the most widely used in image steganography research. Embedding messages using the spatial domain is relatively simple because it can be done directly by modifying the image's pixel values. The weakness of the spatial domain is that the message will be more fragile if there is distortion in the stego image. The advantage of this method is that it has a relatively better imperceptible value, even with a large message capacity. This domain is mostly used in steganography, whose goals are imperceptible and payload. This is evidenced by the data presented in Table 6. Several studies indeed claim to have a large capacity with methods in the transform domain, but when investigated, these methods are indicated to use non-blind extraction. Some of the methods in the spatial domain that are widely popular and widely developed are:

3.2.1.1. Least Significant Bit (LSB). The LSB method is the simplest method that was first developed for image steganography. This method was initially inspired by research [6], then developed in the 90s, and is still being developed today. Steganography based on LSB will be very fragile in case of image manipulation or distortion, such as lossy compression [209]. Another disadvantage of the traditional LSB method is that it is easy to guess when a stego image has been detected. This method is not resistant to steganalysis attacks such as Pair of Value (PoV), RS attack, Chi-square attack, sample pair analysis (SPA), weighted stego analysis, and structural steganalysis [140]. But this method is still very popular because it has advantages such as low complexity and relatively large embedding capacity with good imperceptibility when measured by PSNR [24]. Some of the developments of the LSB method are inverted LSB [93,210,211], LSB matching [49,53,54,123], LSB matching revisited [58,95,212], bit flip [74], combination with adaptive methods such as edge detection [25,46,77], and in combination with artificial intelligence or machine learning [59,213]. In addition, the LSB method can also be combined with transformation domain methods when embedding messages. The development of the LSB method is certainly much more secure than the traditional LSB method because it is more resistant to specific steganalysis attacks and has increased imperceptibility and capacity. Besides that, it can also be developed into a steganographic method that is reversible.

3.2.1.2. Pixel Value Differencing (PVD). The PVD method was initially developed by Wu and Tsai [214]. The idea is to insert a secret message according to the results of quantizing the difference in the adjacent pixel values. By default, the difference is calculated from the value of two pixels in the horizontal direction. This method is more secure than LSB because it is resistant to RS analysis attacks and has a larger capacity than LSB. Generally, the PVD-based method has better imperceptibility than LSB based on HVS measuring tools such as SSIM. It has a histogram that is more similar to the original image histogram [24]. But the PVD standard is weak against PDH attacks. To improve security, imperceptibility, and payload, PVD was developed with several methods, such as modulus function (MF) and multiple or adaptive directional blocks, such as [31,35,75,100], etc.

3.2.1.3. Exploiting Modification Directions (EMD). The EMD method was first introduced in research [215], where $2n + 1$ secret messages are embedded in n cover pixels, in this way the distortion range value is ± 1 . The way of embedding EMD is also different from LSB and PVD and is claimed to have high embedding efficiency. The workings of this method is the use of a specific basis to determine local variations in the intensity of image pixels. So that a greater benefit is obtained in the image area that has a high texture. Thus the EMD method will also produce embedding results according to the HVS principle so that it has better imperceptibility and security. By default EMD has an embedding capacity of 1.16 BPP with $n=2$, and the capacity will decrease if there are n additions [20]. Several studies have since developed EMD to increase payload capacity such as [97,121,216]. In addition, the EMD method can also be developed for reversible data hiding [58].

3.2.1.4. Histogram Shifting (HS). The histogram shifting method in the spatial domain was first introduced by Ni et al. [217]. The idea is to set the lowest bin histogram and highest histogram bin before shifting the histogram. The shift is done in one step, and each cover image's message size differs depending on the lowest bin histogram, especially the highest bin histogram. By using the cover image method used is lossless or reversible. This is undoubtedly useful for some instances that require complete reversibility of the cover image. However, the embedding capacity of this method is generally relatively smaller than that of the non-reversible data hiding method. In its development, this method can also be applied in the transform domain (IWT) [218] based on error prediction [219]. In the latest research, this method performs steganography combined with image enhancement methods [85]. While in research [51], The HS method is combined with the pixel mapping method to increase the imperceptibility to more than 50dB, and the capacity is around 1.5BPP. In research [89], The HS method is combined with several other methods, such as RIWT, Laplacian pyramid, and deep learning to develop robust watermarking.

3.2.1.5. Difference Expansion (DE). The DE method was first introduced in research [220] in the spatial domain and developed in research [221] to create a reversible watermark. At the beginning of the era, the gap between steganography and watermarking was not that far. Still, nowadays, the level of robustness of watermarking and security in steganography is very different. In general, how the DE method works refer to the difference in pixel pairs for embedding secret data. These value differences are expanded using other techniques to expand the range of differences and then pinned based on that range. The DE method has been developed more for reversible data hiding. To improve performance, DE is combined with a modulus function like research [50], modified single pixel on DE with superpixel then adjusted to maximize message capacity [118], and the combination of interpolation and DE with two embedding layers to produce a large capacity [91].

3.2.1.6. Pixel Value Ordering (PVO). The PVO method is also a reversible steganographic using a histogram modification technique. The idea of this method was first proposed in Li et al. [222]. PVO works by dividing the image into blocks of a fixed size, not overlapping, and then each block is sorted sequentially. PVO predicts the smallest pixel with the second smallest value and the largest pixel with the second largest pixel. Then the first prediction error is calculated by the difference between the original value and the second smallest pixel value of the block. The original value calculates the second prediction error with the largest pixel value of the block. In the end, this scheme will generate a histogram of prediction errors and expand the peak bins '1' and '-1', further shifting all bins greater than '1' or smaller than '-1'. PVO will provide high imperceptibility but the effect on a more limited capacity. So that in its development, PVO was developed to increase its capacity with several approaches such as uniform partitioning size, variable partitioning size, adaptive embedding, pairwise embedding, pixel-based PVO, and others [223].

3.2.2. Transform Domain

Steganography in the transform domain embeds the frequency domain. Before embedding the spatial domain, it is transformed into the frequency domain. After the embedding is complete, an inverted transform is performed to obtain a stego image. Likewise, the extraction process must be carried out in the frequency domain. The embedding process in the frequency domain generally makes the embedded message more robust and/or imperceptible and/or secure. But at the beginning of the development of steganography, the transform domain was not very popular because, in this domain, it created a larger distortion effect even though the message would be more robust when passing through a lossy channel. This is because the embedded bits are spread over a wider area of the image, so they are more widely applied for robust watermarking. However, the transform domain is still being developed in the latest steganographic methods because it is relatively more challenging and has the potential to produce secure and robust steganography. Below are some transformations that are widely used in image steganography.

3.2.2.1. Discrete Cosine Transform (DCT). DCT is a domain transformation that was first introduced in Ahmed et al. [224]. DCT is one of the most popular transformations in data hiding science for watermarking and steganography. This is because DCT is the standard transformation used for JPEG compression [225], and JPEG is the standard compression method which most widely used in various digital image transaction processes. DCT works by transforming the spatial domain of the image into two kinds of coefficients one is the DC coefficient, and the other is the AC coefficient. This DC coefficient is the most essential part containing the image's core information. In data science, hiding DCT was initially implemented more in image watermarking. The message bits embedded in the DCT transformation will be spread over a wider area, giving rise to a robust effect. As a side effect, the distortion that occurs in the cover image is larger, although the embedding capacity is smaller than the various methods in the spatial domain. In addition, it makes reversible steganography more complicated. However, the latest DCT steganography method has developed into very robust steganography, such as research [226,227], even developed on reversible data hiding [228,229].

3.2.2.2. Discrete Wavelet Transform (DWT). DWT is a transformation that is no less popular than DCT. DWT is also used as the JPEG 2000 compression standard [230], thus making this method also quite widely used in steganographic research or data hiding. DWT transforms the image from the spatial domain with a low pass filter and a high pass filter. The result is that the image will consist

of four subbands, namely LL, HL, LH, and HH. This transformation has many developments, such as the integer wavelet transform (IWT). In some steganographic research such as [29,42,83,104], IWT was chosen because using integer values improves the extraction process and reduces the possibility of missing message bits. Wavelet transform has also been developed into its derivative transformations, such as bi-orthogonal wavelet transform [55], Dual-tree complex wavelet transform [37,117], and wavelet package transform [34]. Wavelet filters are also widely used in secure steganography methods such as WOW, UNIWARD, and HILL.

3.2.2.3. Catalan Transform (CT). CT is one of the invertible transformations in the sequence of integers associated with Catalan numbers. CT can give effect characteristics to general sequences such as Fibonacci and Jacobsthal [231]. The Catalan transformation is not new but less popular than the DCT and DWT transformations. Catalan transform has not been widely used in image steganography and was only implemented in 2021 in research [36,232].

3.2.2.4. Framelet Transform (FT). FT is built to fix two problems that occur in wavelets. First, a slight shift in the input signal causes significant variations in the wavelet energy distribution at different scales. Both diagonal feature selectivity on the wavelet is not good. FT is described by two linear operators, namely subdivision and transition. Subdivision operators are tasked with reconstructing, refining, and predicting data to a higher level. The transition operator decomposes and coarsens the data frequency to a lower resolution level. FT is built with three functions: the scaling function as a lowpass filter and two wavelet functions as a highpass filter. These functions make three fundamental properties: sparsity, stability, and perfect reconstruction. FT will produce subbands, and there is redundancy between subbands. Thus, when there is a change in the coefficient in one sub-band, it can be compensated by another sub-band. This makes FT less sensitive to coefficient modification, so FT is suitable for use in steganography because it can accommodate more payloads but does not much affect the quality of the stego image [32]. Some steganographic research that uses FT as one of its bases are [32,233,234].

3.3. Based on Distortion Function

The steganographic method based on the distortion function has a different approach in producing a stego image. The previous steganography method is designed to retain the original cover model completely or roughly. While distortion function-based steganography aims to produce DM (distortion-minimization) with optimal insertion that operates at the limit of theoretical distortion rate and practical coding scheme. In this case, the additive distortion function is more widely used because it can be expressed as the sum of the embedding costs, which measures security by modifying the enclosure element accordingly. The main mission of these methods is three rules. The first is the complexity rule, which assigns low costs to coefficients in complex areas and high costs to coefficients in smooth areas of the image. The second important rule is the spreading rule. It is a priority that the two neighboring elements should not differ much. In short, elements with high priority must propagate high rankings to their environment, and vice versa for elements with low priority. The last is the cluster rule, where this rule conjugates that it is better to make cluster embedding modifications rather than scatter [207]. The method that is built based on the distortion function mostly uses convolutional codes based on STCs. The following are some of the widely developed methods with distortion functions:

3.3.1. Highly Undetectable steGO (HUGO)

The HUGO method was first introduced in research [203]. This method is built using different techniques with LSB, PVD, EMD,

and histogram-based methods. This method works in the spatial domain that utilizes the high dimensions of the image. To design a steganographic method that is very secure and difficult to detect by steganalysis tools. This method is built based on the features contained in SPAM [154] and the embedding process based on syndrome-trellis codes (STCs). Based on the test results on the BOWS2 database, the HUGO method is proven to have better security than LSB matching. With the same detection error rate of 40%, HUGO can load 0.3BPP while LSB Matching is only 0.04BPP.

3.3.2. Wavelet Obtained Weights (WOW)

The WOW method was first published in Holub and Fridrich [204]. WOW is designed using an additive distortion function in the spatial domain but uses a different strategy than HUGO. WOW uses a set of directional high-pass filters based on 2D wavelet filters to obtain directional residuals related to the predictability of pixels in a given direction. Unlike HUGO, WOW uses SRM's high-dimensional features to conduct training. The STCs method is also used in the WOW algorithm for embedding. Features in WOW can limit embedding changes to textures while avoiding "clean" edges, thereby increasing the safety of steganography. Based on the test results on the BOSSbase database, the WOW method successfully outperformed the HUGO method.

3.3.3. Universal Wavelet Relative Distortion (UNIWARD)

The UNIWARD method is a distortion function-based steganography method that can work in the spatial domain (S-UNIWARD) and the JPEG domain (J-UNIWARD). This method was introduced in Holub et al. [205] as the development of the HUGO and WOW methods. The distortion function for insertion is calculated as the relative change of the coefficients in the directional filter bank decomposition of the cover image. Directions force embedding changes to parts of the cover object that are difficult to model in multiple directions, such as textures or noisy areas, while avoiding smooth areas or clean edges. The high dimensional feature used in S-UNIWARD is a combination of content-selective residual (CSR) and SRM, while J-UNIWARD uses SRM feature development, namely SRM Quantization 1 (SRMQ1) and JPEG rich model (JRM). Based on the experimental results, the UNIWARD method performs better against steganalysis attacks than HUGO and WOW.

3.3.4. High-pass, Low-pass, and Low-pass (HILL)

Steganographic methods such as WOW and S-UNIWARD can exploit more pixels in the texture area to hide messages. However, the few pixels in the texture area that may be suitable for conveying a message come at a high cost. Intuitively, distortion in pixels due to embedding should be more secure than in smooth areas. The HILL method was first proposed in research [206] to solve the problem of embedding distortion by minimizing the cost function. The novelty of the HILL method is the use of a new cost function with three filters consisting of one high-pass filter and two low-pass filters. Based on testing using the SRM feature and mapping co-occurrence using non-uniform quantization (MCNUQ) [235], the HILL method outperforms the MOW and S-UNIWARD methods.

3.3.5. Steganographic Polar Codes (SPC)

SPC is one of the newest alternatives to STCs. SPC was first published in Li et al. [188]. The SPC method is built with a successive cancellation list (SCL) decoding algorithm to minimize additive distortion in steganography. SPC performance has been proven to be better than STCs overall, with low embedding complexity in image objects. SPC has been verified to construct a steganographic code practically based on the progress of polar coding/decoding. Tests carried out by applying SPC to HILL, UERD, S-UNIWARD, and J-UNIWARD have the advantage of SRM, DCTR, and GFR steganalysis.

3.4. Steganography Classification Summary

Based on the data presented in section 3, we classify the steganographic method based on goals, domains, models, and reversibility. Based on the goals, it is clear that the steganography method has four goals: imperceptible, payload, secure and robust. But in general, steganographic research has at least two goals: imperceptibility and payload or security and robustness. The goal imperceptibility and payload generally use a statistical embedding model in the spatial domain, although some methods also use the transform domain (see Table 6). Several techniques have also been developed in the recent spatial domain steganography, such as recursive information hiding schemes, enhanced adaptive data hiding, hybrid methods, and embedding based on pixel intensity [20].

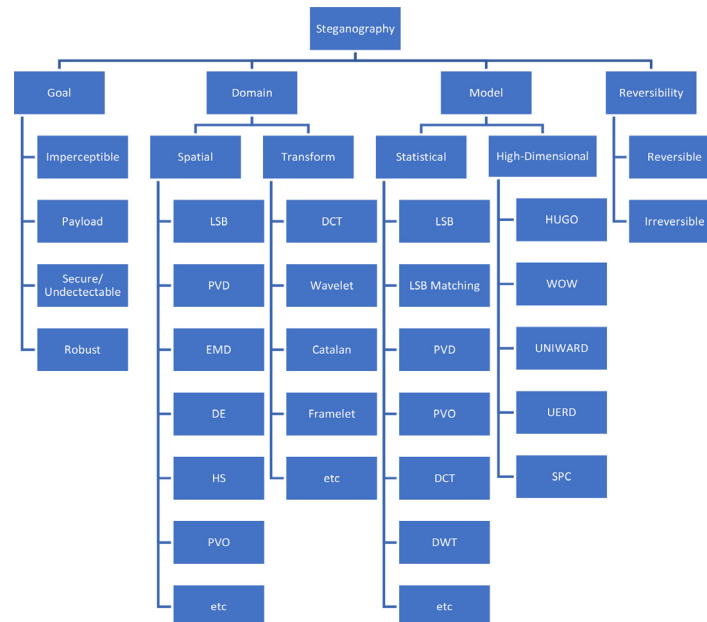


Fig. 8. Image Steganography Classification Summary.

Steganography methods can be reversible or irreversible. Reversible means that the cover image and the message are lossless. So reversible steganography is widely used in applications that require a cover image that can be fully recovered. While the irreversible, the cover image is lossy. In some of the latest methods, statistical models are considered less secure, and high-dimensional models are starting to be widely used. The statistical model itself is not only used in the spatial domain but also in the transform domain. Steganography with high-dimensional models is generally more secure and robust but has a relatively small payload. In steganography methods that use high-dimensional models, they typically use universal steganalysis for security testing. For an overview of the classification of the steganography method, see Fig. 8.

4. Developments in Image Steganography

Digital data hiding methods have evolved since the 90s when the gap between steganography and watermarking is not that far away. Along with technology development, new ideas and approaches are increasingly varied, so novice researchers find it difficult to distinguish between one data hiding technique and another. So this section discusses steganographic derivative methods, sophisticated steganographic methods, and special-purpose steganography.

4.1. Coverless Image Steganography (CIS)

CIS is a new idea and concept of steganography. CIS is different from general image steganography which has been discussed above. Another name for CIS is steganography without embedding (SWE). Undeniably, no matter how good a steganography method is, it will produce distortion in the cover image. The distortion that occurs will make it possible for the steganalysis tool to detect the message. All image steganography methods cannot eliminate this problem [236]. The CIS idea was first proposed in Zhou et al. [237]. This is done by utilizing features in the cover image, such as pixel brightness, color, texture, edges, contours, and high-level semantics, to “generate” a cryptographic cover. This cryptographic cover generates some relationship between this feature information and confidential information for hidden information with an appropriate feature description. So CIS does not mean the carrier is not

used but is not modified. CIS/SWE has a different concept from the steganography proposed in several studies, such as [92,238–241]. The basic idea of CIS is to analyze carrier attributes and map them to confidential information according to certain rules based on the characteristics of the attributes. In this way, the carrier can directly represent the confidential information, see Fig. 9. In short, the main contribution of unclogged image steganography is that covert communication can be realized without modifying the stego-image so that confidential information cannot be detected by the steganalysis tool.

In the development of the CIS method is generally developed by increasing the accuracy of reading features. The goal is to make image retrieval more appropriate, accurate, and precise, as in research [242] which uses the histogram of gradient, [243] use partial duplication, [244] use the generative adversarial network (GAN), [245] use LDA topic classification, and [246] using dense net features and DWT sequence mapping, [247] using optical mark recognition and machine learning.

4.2. Generative Adversarial Network (GAN) based Image Steganography

The GAN-based steganography method is motivated by the development of the convolutional neural network (CNN) based steganalysis method. The increasingly robust CNN-based steganalysis poses a threat to the security of steganography. GAN-based methods generally use adversarial noise to improve cover quality to obtain an image structure that is in accordance with the information to be embedded [183]. Currently, several GAN-based steganographic methods are being followed. MiPOD method [126] is one of the pioneers of GAN-based steganography. Followed by methods such as ISGAN [71], which focuses on increasing invisibility. AEN [187], and HIGAN [189] focus on minimizing distortion and increasing security. Another GAN method was proposed based on cross feedback by Li et al. [124] to improve security which has been tested with several steganalysis such as SRM, maxSRMd2, Xu-Net, Ye-Net, Yedrouj-Net, and Deng-Net. Another GAN method is proposed [125] using sparse cover enhancement (SPS-ENH), which increases the security of secret messages and minimizes distortion. This method was tested with SRM, SRnet, and YeNet steganalysis.

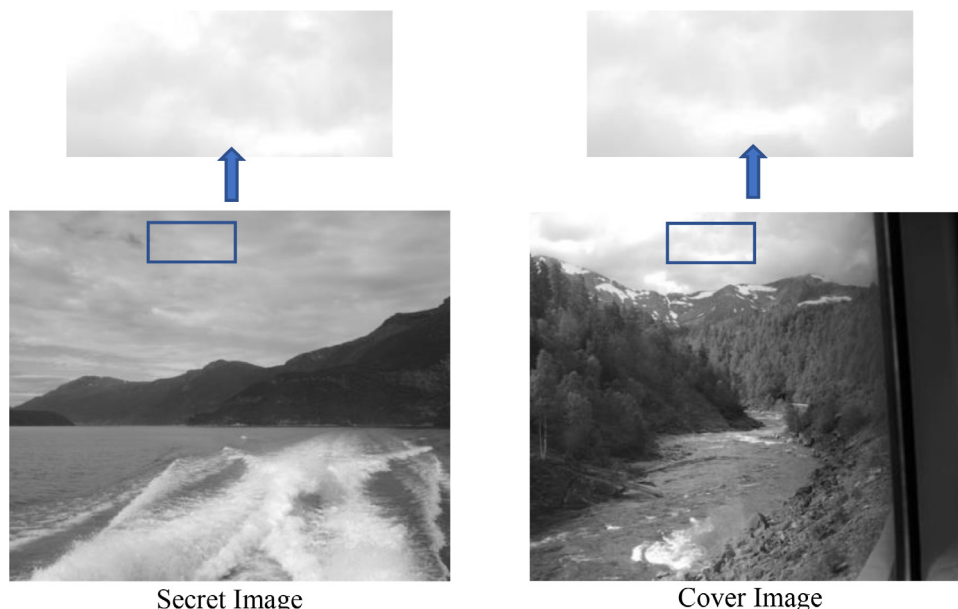


Fig. 9. A Cover image and a secret image contain one identical patch; information contained in secret images can be transferred through cover images.

4.3. Cover Selection Strategy for Image Steganography

The cover selection method in image steganography is actually not new. This method was first proposed in research [248]. This method aims to maximize the embedding rate and minimize detectability by choosing a cover based on the features or properties of the image with the embedded secret message bits. At the beginning of the development of this method, standard measuring tools were used to obtain image properties such as changeable coefficients, JPEG quality factor, number of modifications, MSE, prediction error, Watson metric, and SSIM. With this method, it can be proven that more effective embedding is obtained. There are not many steganographic studies like this, but this technique continues to develop and is applied in some of the latest steganographic methods, such as [249–254]. This method uses machine learning and deep learning to study features and select cover images. The latest research shows even combining selection techniques and stego generation using GAN [255].

4.4. Special Purpose Steganography

From various steganographic research, there are also methods designed for special purposes. For example, in research [256] developed a new idea known as multichannel steganography, in which a sender can deliver private data to several receivers using the same cover image. This technique also includes priority levels, allowing high-priority recipients to adjust the permission for lower-priority users. Another method [257] proposes an iterative data hiding system in which the distortion created by data embedding remains constant regardless of how many times the embedding process is performed. This is accomplished by inferring the theoretical alteration probabilities of the cover image's components. When further data with inferred probability is embedded into the cover image, the changed trace of the current embedding is replaced by the last embedding.

4.5. Other methods have similarities/derivatives of steganography/data hiding

Aside from the methods discussed above, there are numerous more similar to or derivatives of steganography or data hiding. Among these techniques are secret image sharing (SIS) and visually meaningful image encryption (VMIE). Actually, SIS is not a novel approach, and it is more closely related to cryptography than steganography because it was first used in visual cryptography (VC). VC is another method for securely encrypting secret image (SI) information. The original image (called secret image) is encrypted in VC and converted into a separate image known as a share. This amount is divided among the participants. A single share cannot represent SI. The original image is captured when these shares are stacked together. Pixel expansion is a challenge for traditional VC systems. The SI is converted into a bigger size in this case. The visual secret sharing scheme (VSS) technique addresses this issue further. VSS is a method for securely dividing the SI into pieces. The advantage of this approach is that HVS can decode the SI, and no cryptographic computations or expertise are required [258].

VMIE is an encryption technique using information-hiding techniques. The original image is first encrypted to SI. Then commonly, the transformation domain is used in the cover image so that later it can be used to cover the SI. Then the reverse transformation is carried out after the SI is embedded. As a result, SI will be visually meaningful [259–261]. We need to discuss this briefly because VSS and VMIE have something in common with steganography, namely, the embedding process. Assessment in VMIE also has similarities with steganography, such as imperceptibility and robustness. Still,

in VMIE, it is necessary to add other cryptographic assessments, such as key sensitivity, key space, etc.

5. Image Steganography Dataset

Image dataset is an essential part of testing the proposed steganographic method. Research in the field of good steganography generally contributes to the method rather than the problem. It can be seen in Table 1 almost all studies use public datasets. As for those who use private datasets, they also use public datasets to facilitate the comparison process. With a public dataset, the comparison process will be easier without replicating the method. The following is a dataset that is widely used in the steganographic method:

5.1. Standard / Classic Image Dataset

Standard or classic image datasets that are widely applied to image steganography are Lena, Baboon, Pepper, and so on. This dataset is actually not specifically for steganography research, but for various image processing research. This dataset can be downloaded on the USC SIPI Volume 3: Miscellaneous page (<https://sipi.usc.edu/database/database.php?volume=misc>) and standard image test on url (https://www.imageprocessingplace.com/root_files_V3/image_databases.htm). This dataset is most widely used in the steganographic method with goal imperceptibility and payload capacity. But the number of these datasets is limited and not large, so it is quite difficult to apply steganographic methods based on machine learning, deep learning, or GAN.

5.2. BossBase Image Dataset

The BossBase dataset was first popularized in [262] in 2011. There are 10,000 images in this dataset with PGM extension. The image size is 512×512 pixels with a depth of 8-bits. This image can be downloaded on the website <http://dde.binghamton.edu/download/>. Currently, the BossBase dataset has version 1.01. This dataset is a dataset designed for steganalysis and steganography. This dataset is most widely developed for steganographic methods that focus on developing security or undetectability from machine-based or deep-learning universal steganalysis attacks.

5.3. BOWS2 Image Dataset

The BOWS2 dataset was first published in Furon and Bas [263] and was designed for watermarking. But this dataset is quite popularly used in steganographic research such as [29,33,35,45,102,179,213], dst. This is because of the resemblance to watermarking. This dataset can be downloaded on the <http://bows2.ec-lille.fr/>.

5.4. ALASKA v2 Image Dataset

One of the newest and largest datasets in image steganography is the ALASKA v2 image dataset. This dataset is designed for steganalysis and steganography research. The size of this dataset reaches 32.27 GiB, which consists of a cover image and a stego image embedded with three algorithms, namely J-MiPOD [127], J-UNIWARD [205], dan UERD [185]. The average payload embedded in the stego cover is 0.4 bpnzAC. This dataset was first published in [264] and can be downloaded on the url <https://www.kaggle.com/competitions/alaska2-image-steganalysis/>. Like the BossBase dataset, the ALASKA v2 dataset is widely used in steganographic research that favors security or undetectability.

5.5. IStego-100 K Image Dataset

This dataset was first published on Yang et al. [265]. This dataset consists of 104,052 cover images and 104,052 stego images, so the total is 208,104 images. The stego image in this dataset is designed using the J-UNIWARD, nsF5 and UERD algorithm with a payload of 0.1-0.4 bpnzAC at random. All images have dimensions of 1024×1024 pixels, of which 200,000 are training images and the rest are test images. Like BossBase and ALASKA, this dataset is designed for universal steganalysis and safety testing in steganographic research. This dataset can be downloaded via url <https://github.com/YangziITHU/IStego100K>.

5.6. RGB-BMP Steganalysis Dataset

RGB-BMP Steganalysis Dataset consists of 3000 images with a size of 512×512 pixels which can be used for steganography, steganalysis, and image processing researches. The number of this dataset is not as much as ALASKA, BossBase, or IStego. However, this dataset can still be used as an alternative. One of the studies using this dataset is [36]. This dataset can be downloaded at <https://data.mendeley.com/datasets/sp4g8h7v8k/1>.

5.7. Other Image Dataset

Not all datasets are specifically designed for steganography or steganalysis, but these datasets are used in steganographic research. Some of them are the UCID image dataset first published in Schaefer and Stich [266], and some of the studies using this dataset are [26,43,89,110,115,119,121,134,252]. The UCID dataset can be downloaded at the link <http://jasoncantarella.com/downloads/ucid.v2.tar.gz>. Corel Image dataset is used in several researches such as [41,87,267] and can be downloaded on <https://sites.google.com/site/dctresearch/Home/content-based-image-retrieval>. Kodak Image Dataset used in research [85] can be downloaded at <http://r0k.us/graphics/kodak/>. Washington image database can be downloaded on the page <http://www.cs.washington.edu/research/imagedatabase>. This dataset is used in several steganographic research such as [32,253]. Next is the McGill Database, which consists of 1034 uncompressed images with dimensions 768 × 576 or 576 × 768 pixels. The McGill dataset was first published in [268] and used in research [121,123].

6. Summary and Future Steganography

Steganography in digital images has been developed in the last three decades as a technique for securing secret messages by hiding them on the cover media. At the beginning of the development of the steganography method, the gap with watermarking was not too far because the level of robustness of watermarking nor the level of security or imperceptibility of steganography was not too significant. The most distinguishing thing is the message extraction process. In steganography, it should use a blind detector, while in watermarking, it can use a blind or non-blind detector. The LSB method is a spatial domain method, the first to be developed in image steganography. Furthermore, many new methods are proposed such as PVD, EMD, HS, PVO, DE, etc., even in the transform domain. In the domain transform, the insertion is carried out after the transformation process to the frequency domain, which is then carried out by the invert transform process to obtain a stego image. These methods can produce imperceptibility qualities that are in accordance with HVS, and some methods can be reversible. An important note regarding the characteristics of reversible and irreversible is a research cannot compare the performance of reversible and irreversible methods. Because apart from having different characteristics, the complexity of the method is also differ-

ent. This is because reversibility requires more effort and needs to sacrifice other aspects, such as reducing the payload.

The novelty of the steganography method can be proven by improving at least one aspect, especially the security aspect, as evidenced by the assessment tools described above. Four aspects are used to measure stego image quality: imperceptibility, payload capacity, security/undetectability, and robustness. But the robustness aspect is generally not very important unless the steganography is designed to pass through lossy channels such as social media networks. These aspects have trade-offs, especially in the payload aspect. The larger the payload will affect the amount of distortion, thereby reducing the quality of imperceptibility and security. This has become one of the challenges in image steganography. Another challenge is the advancement of steganalysis technology. Currently, steganalysis is not only designed to detect specific methods but can detect methods universally. Steganalysis is designed with a learning system on image features and convolution methods. This forces to improve the security performance of the steganographic method. Secure steganography has begun to be proposed not only with statistical models, but with distortion functions based on high-dimensional models. It doesn't stop here, and the latest steganography method is even developed with a cover selection strategy GAN, even a coverless technique to improve the security of sending messages. However, until now, no steganography technique has a large payload with good imperceptibility, security, and robustness. Generally, the steganographic method excels in one or two aspects. In the future, distortion-based steganography, feature extraction, deep learning, and adversarial network methods will be more popularly developed. Combining the feature selection model to modify or select the optimal cover will be a very challenging direction of steganographic research. Besides that, there will be more steganographic methods made with special functions, as well as hybrid techniques and their derivatives. But the most challenging thing is the parallel development of the four main aspects and practical implementation on social media or popular messaging applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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